

**LABORATORY ACTIVITY NO. 2.4.10****Creating and Using Databases**

The lab activity is designed to teach students the fundamental database management skills using Apache OpenOffice Base. This exercise will walk them through the process of setting up a database environment, which includes installing necessary software, creating a new database, designing and implementing tables with primary and foreign keys, establishing relationships between these tables, and inputting data efficiently using forms.

I. OBJECTIVES

Upon completion of this laboratory exercise, students are expected to and meet the following intended learning outcomes:

Activity Objectives	Intended Learning Outcomes		
	ILO1	ILO2	ILO3
1) successfully install the latest version of Apache OpenOffice and the required 32-bit Java Runtime Environment on a Windows 11 system, then create a new database using OpenOffice Base.	•	•	
2) design and implement database tables within OpenOffice Base, correctly defining primary and foreign keys to ensure data integrity and establish foundational relationships between tables.	•	•	
3) establish and verify relationships between created database tables, and efficiently populate data into these tables using custom forms in OpenOffice Base.	•	•	

Where:

- ILO1 – Exhibit thorough demonstration of the concept
- ILO2 – Being able to perform the activity independently
- ILO3 – Design and evaluate emerging IT solutions

II. DISCUSSION

- **Database Concepts**



- Databases store highly structured data in tables, which consist of columns (fields) and rows (records).
- Database usage involves creation, import/input, storage (data persistence), queries, and reports.

- **Flat File Systems versus Databases**

- Databases offer advantages over flat file systems (like spreadsheets or CSV files) by defining column data types, validating data entry, storing and linking multiple tables in a schema, supporting many users with high transaction rates, being more scalable, and using more sophisticated access controls.

- **Database Structure**

- Relational Database Management Systems (RDBMS) often use Structured Query Language (SQL). Examples include Microsoft SQL Server, Oracle Database, MySQL, and Microsoft Office Access.
- Data is stored efficiently in multiple tables linked by primary and foreign keys.
- A **primary key** is a designated table column with a unique value for each record.
- A **foreign key** is a column that references the primary key value of another table.
- The definition of tables, columns, and relations is called the **database schema**.

- **Constraints**

- Data types for columns act as a basic constraint on what can be entered, preventing "Garbage In, Garbage Out" (GIGO).
- Other constraints include uniqueness (e.g., primary key), validation rules (e.g., cannot be left empty, specific format), and default values.

- **Semi-structured and Unstructured Databases**

- **Structured relational databases** have defined tables and columns with data types.
- **Unstructured data** includes documents, presentations, spreadsheets, and images where the user is not constrained on entry; data is extracted by searching and indexing file content.
- **Semi-structured data** involves documents with metadata or markup that identifies contained values.

- **Document and Key/Value Pair Databases**

- **Document databases** are semi-structured, requiring the database engine to parse each document type, often using XML (eXtensible Markup Language).



- **Key/value pair databases** store object properties without predetermined fields and data types, commonly using JSON (JavaScript Object Notation) and often associated with NoSQL (or Not Only SQL) databases.
- **Data Definition Methods (DDL)**
 - DDL commands modify the structure of the database. Key commands include:
 - CREATE: To create a database or table.
 - ALTER: To alter database properties or table fields (add, remove, modify).
 - DROP: To delete a database or table.
 - CREATE INDEX: To specify that a column is indexed for faster queries.
- **Data Manipulation Methods (DML)**
 - DML commands insert, update, or extract records:
 - INSERT INTO TableName: Adds a new row.
 - UPDATE TableName: Changes values of columns, can be filtered with WHERE.
 - DELETE FROM TableName: Deletes records, can be filtered with WHERE.
 - SELECT: Defines a query to retrieve data.
- **Permissions**
 - Databases use account-based access control systems.
 - Database owners can grant or deny permissions to other user accounts using commands like

GRANT permission TO user and DENY permission TO user. Permission statements are considered DDL commands.
- **Database Access Methods**
 - **Direct/manual access:** Using administrative tools like phpMyAdmin.
 - **Query/report builder:** Provides a GUI to convert user selections into SQL statements.
 - **Programmatic access:** Applications interact using SQL commands or stored procedures.
 - **User interface/utility access:** Applications use databases in the background (e.g., GUIs or forms).
 - Includes backups and data export.
- **Application Architecture Models**



- **One-tier:** User interface and database on the same computer.
- **Two-tier (Client-server):** Client runs application/presentation, server runs database.
- **Three-tier:** Client (presentation), application/business logic, and database are separate tiers.
- **N-tier:** Any multi-tier architecture, more complex than three-tier, potentially including layers like access control/security.

III. MATERIALS

A Windows 11 installed computer system.

IV. PROCEDURE

For this activity, you will execute the provided step-by-step instructions. As you complete each step, capture a screenshot of the resulting output or completed action that directly demonstrates successful execution of that instruction. Paste each screenshot into the designated area within your lab document.

Important Considerations:

- **Software Versions:**
 - The latest stable version of Apache OpenOffice, which is 4.1.15 was released on December 22, 2023.
 - Apache OpenOffice Base, the database component, requires Java to function. The latest versions of OpenOffice do not include a JRE in their installer. Thus, it is highly recommended to install a compatible 32-bit JRE, such as Oracle Java 8 or OpenJDK 8, even if your Windows 11 system is 64-bit, as the Windows version of OpenOffice is 32-bit.
- **Installation Process:**
 - The installation of Apache OpenOffice typically involves downloading the installer from the official website and following a setup wizard.
 - Remember to install the appropriate 32-bit JRE *before* or concurrently with OpenOffice to ensure all database functionalities are available.
- **Security Warning:**
 - Be aware that you will be downloading and installing programs from third-party websites (Java and OpenOffice). While these are reputable sources, always



exercise caution and ensure you are downloading from the official websites to avoid potential risks.

- **Database Concepts:**

- Pay close attention to the concepts of primary and foreign keys and how they are used to establish relationships between tables. This is crucial for maintaining data integrity and building a functional database.
- The lab provides a practical understanding of how forms facilitate data entry and how queries can be used to retrieve and view specific records.

- **Troubleshooting:**

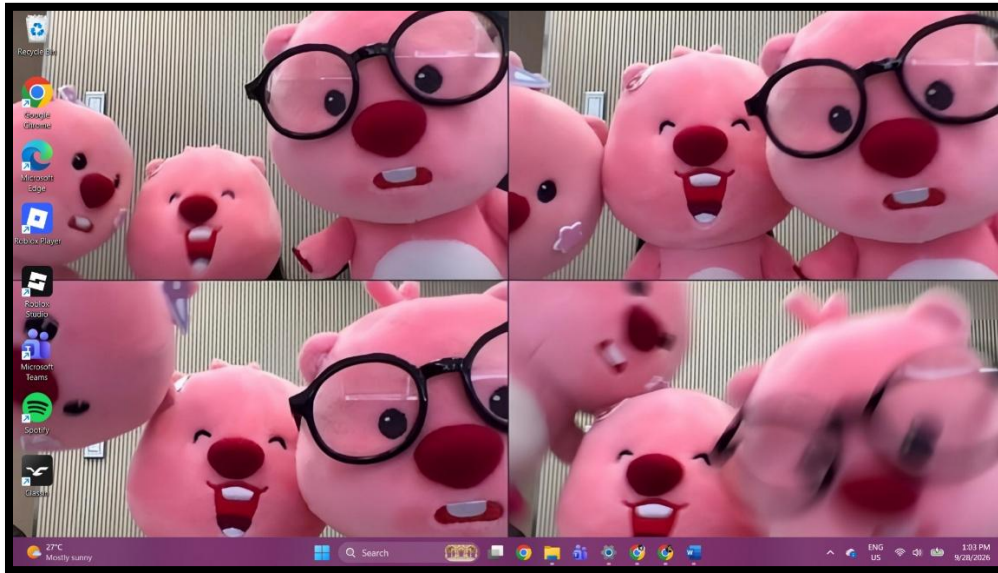
- If you encounter issues with database functionality within OpenOffice Base, it is often related to the Java Runtime Environment installation or its configuration within OpenOffice. Ensure the correct 32-bit JRE is installed and that OpenOffice is configured to use it (Tools > Options > OpenOffice > Java).



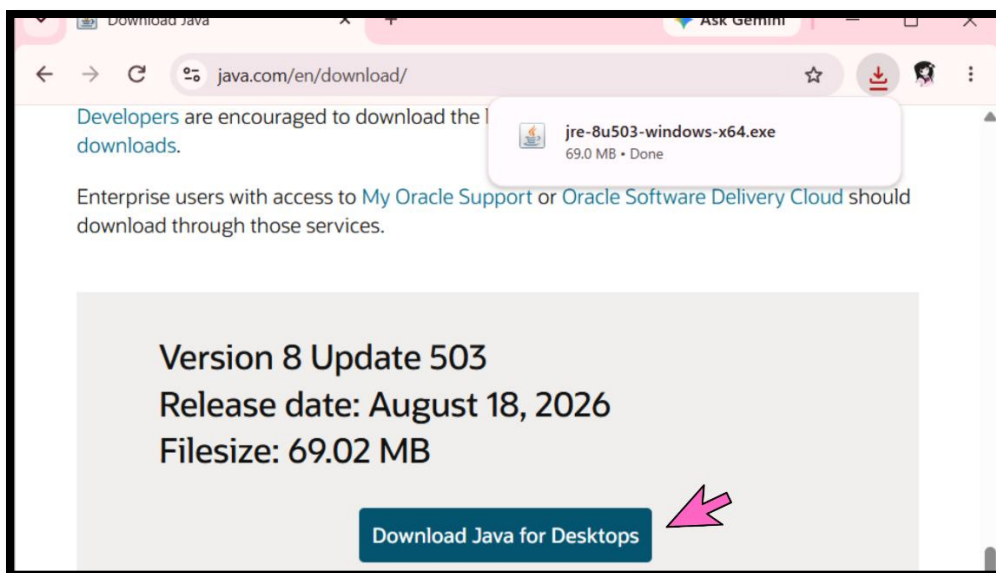
Steps:

Part 1: Installing Apache Open Office

1. If necessary, start your computer and sign in.

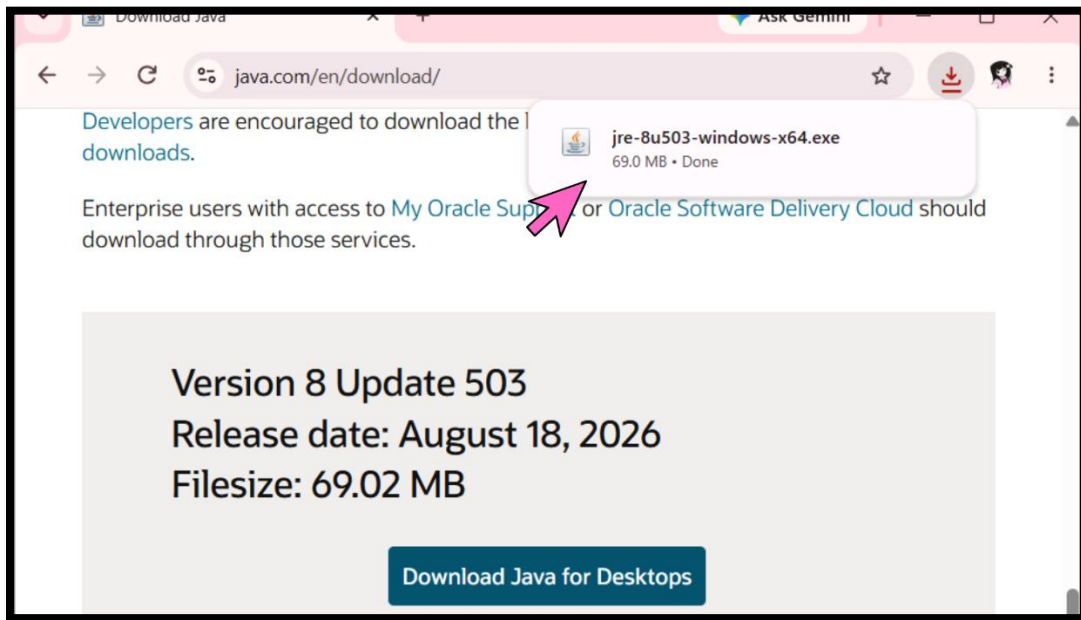


2. Open a browser and navigate to the official Java download page, such as java.com/en/download/. Download the **32-bit version of Java Runtime Environment (JRE) 8** (e.g., `jre-8uXXX-windows-i586.exe` for Oracle Java, or an equivalent OpenJDK 8 32-bit version). Apache OpenOffice for Windows is a 32-bit application and requires a 32-bit JRE, even on a 64-bit Windows system.

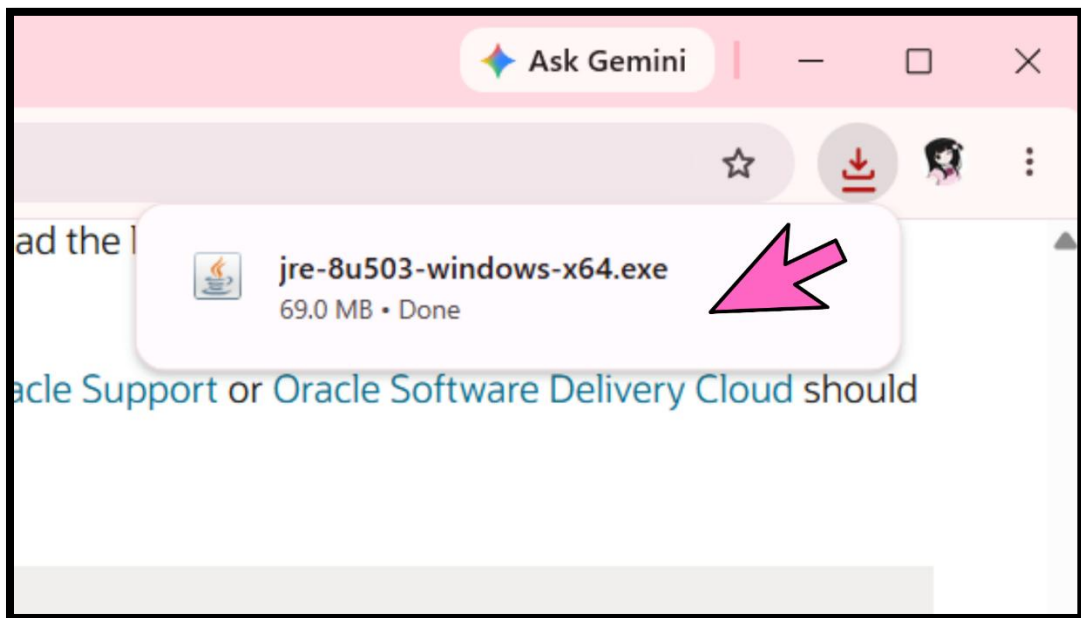




3. If necessary, accept the use of cookies by the site. On the main page, click the appropriate download button.

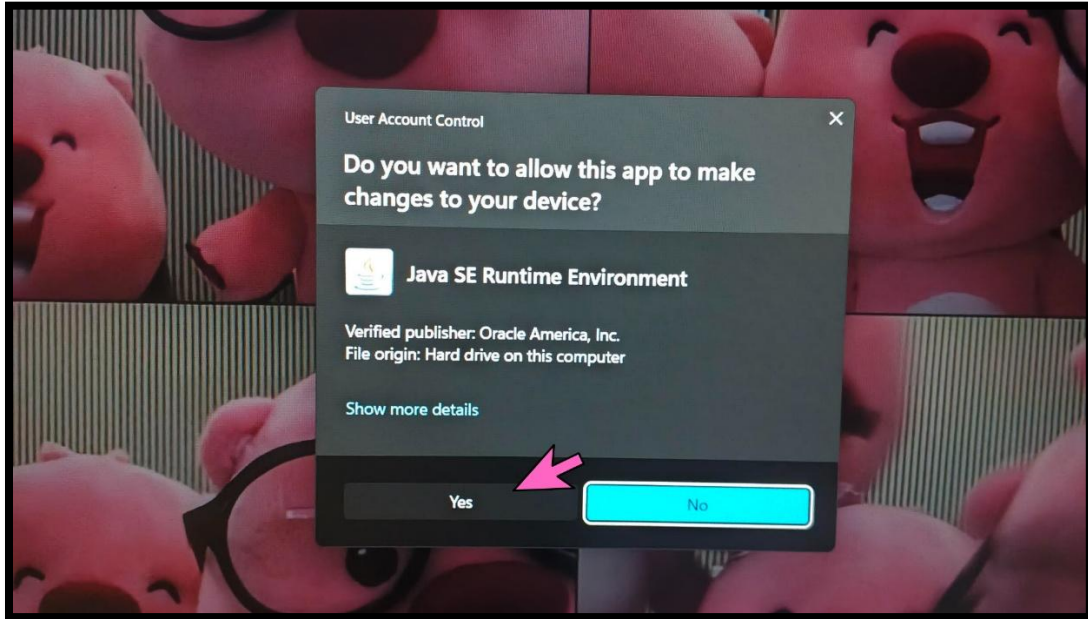


4. In the prompt at the bottom of the window, click **Save**. When the download is complete, in the prompt, click **Run**.

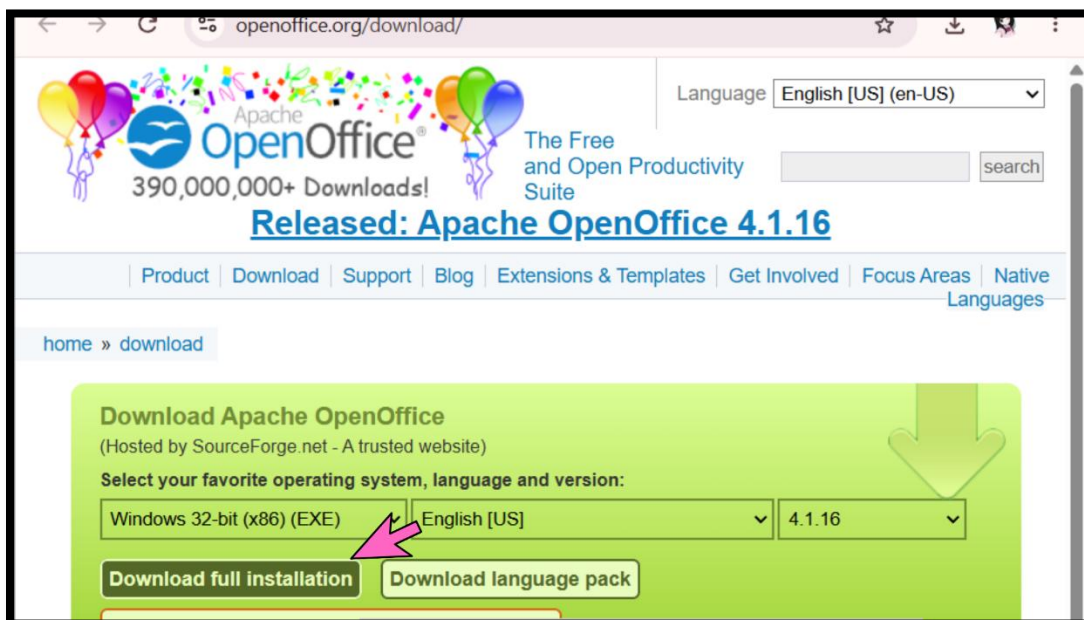




5. Follow the prompts in the Java installer to complete the installation. In the User Account Control window, click **Yes**.

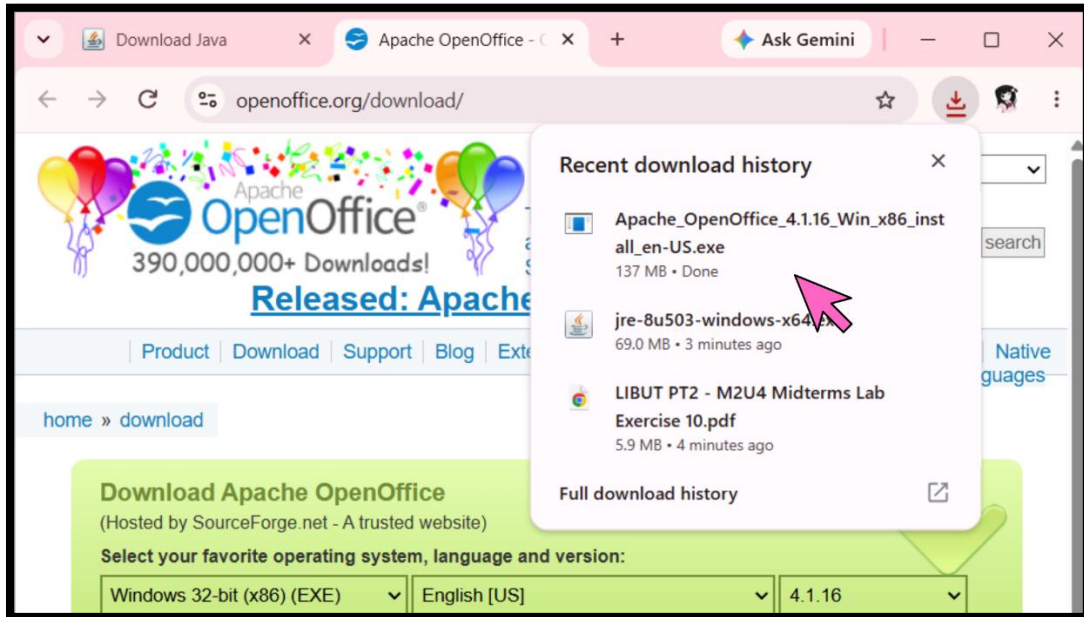


6. Open a browser and navigate to the official Apache OpenOffice download page: www.openoffice.org/download/. Download the latest stable version (e.g., Apache OpenOffice 4.1.15).

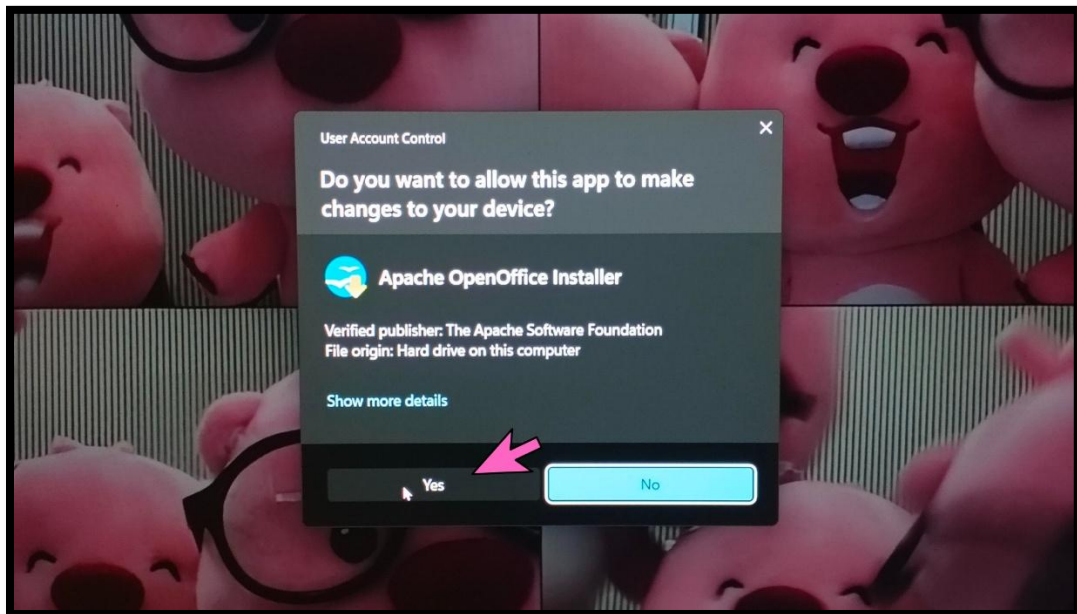




7. In the prompt at the bottom of the window, click **Save**. When the download is complete, in the prompt, click **Run**.

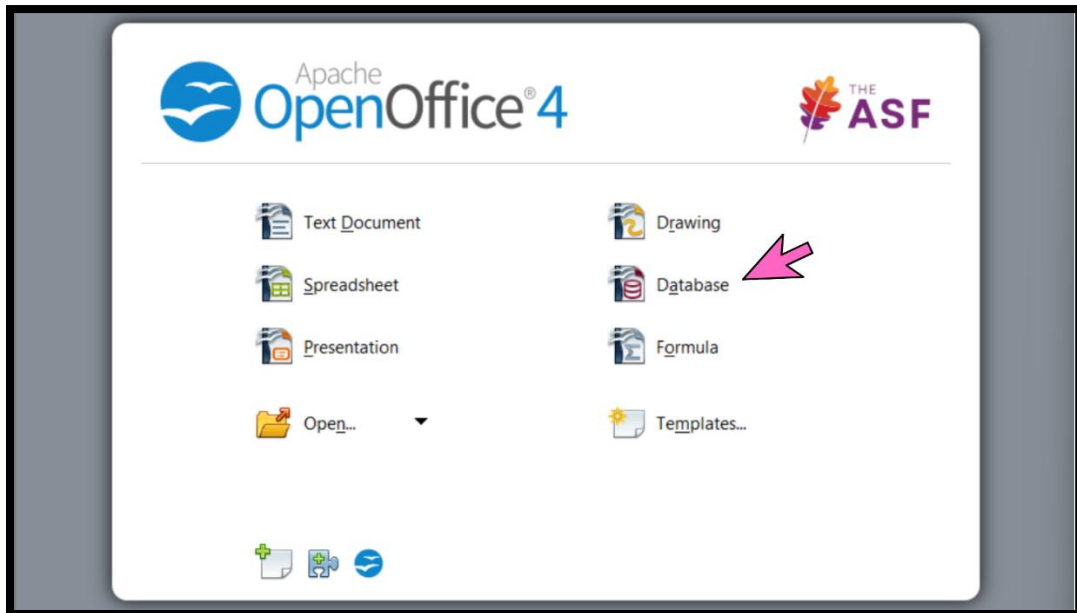


8. In the Apache OpenOffice installer window, click **Next** and follow the prompts. In the User Account Control window, click **Yes**.

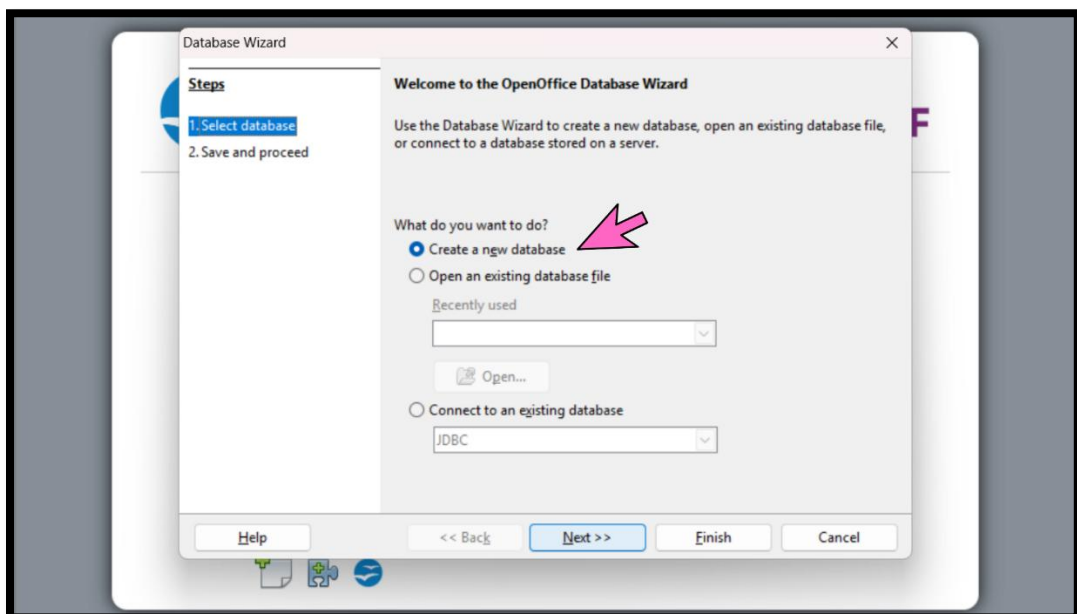




9. Once the installation is complete, click the **Start button** (on the taskbar), type **OpenOffice** in the search bar, and then click **Apache OpenOffice Base** from the search results. Alternatively, if an OpenOffice icon appears on your desktop, you can double-click it and then select **Database** from the application choices.

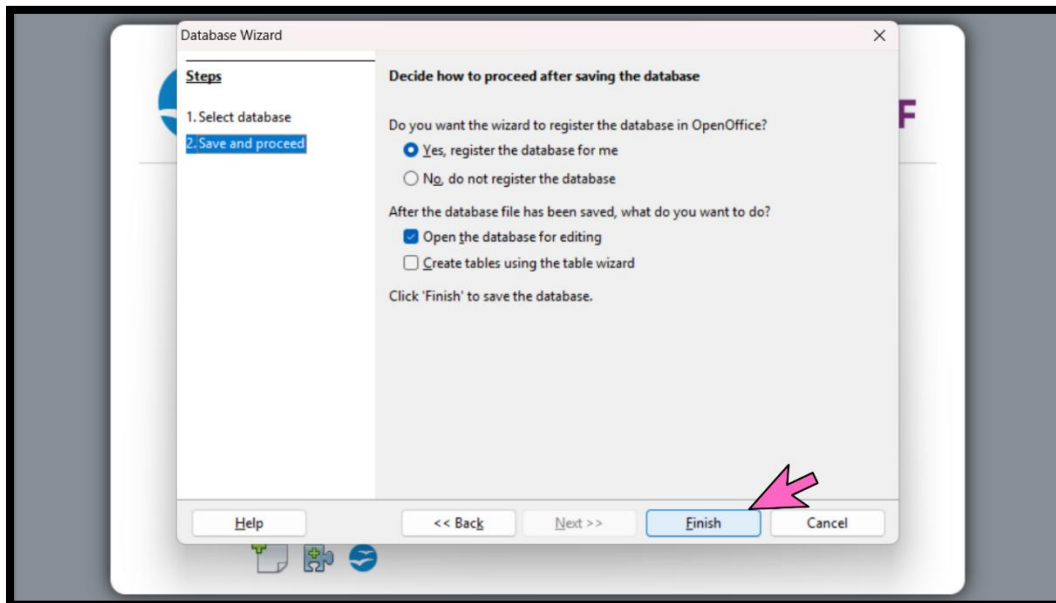


10. In the Database Wizard window, ensure **Create a new database** is selected, and then click **Next**.

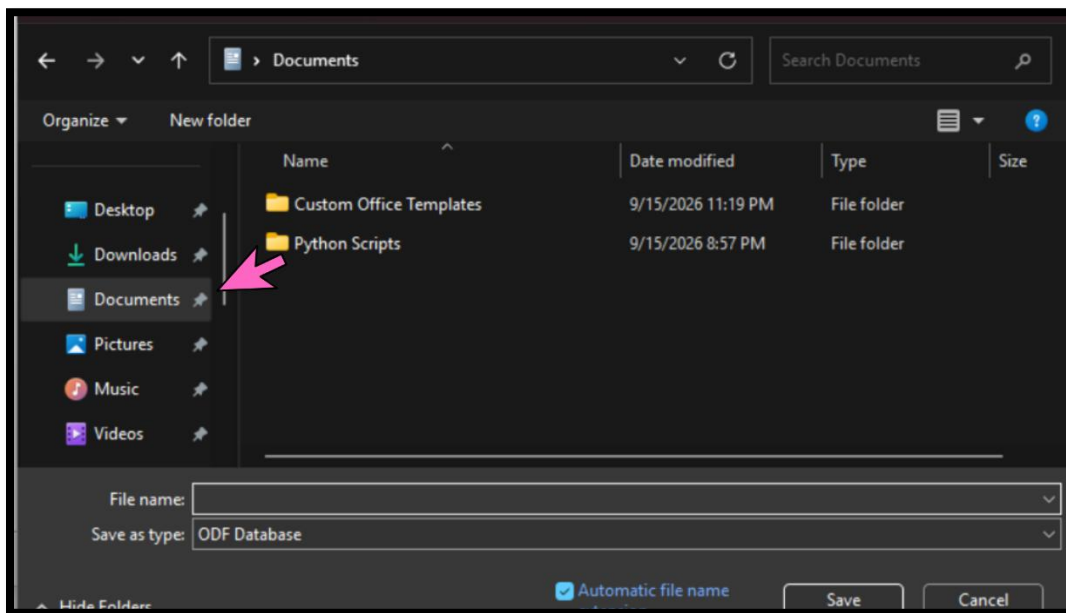




11. Click **Finish**.

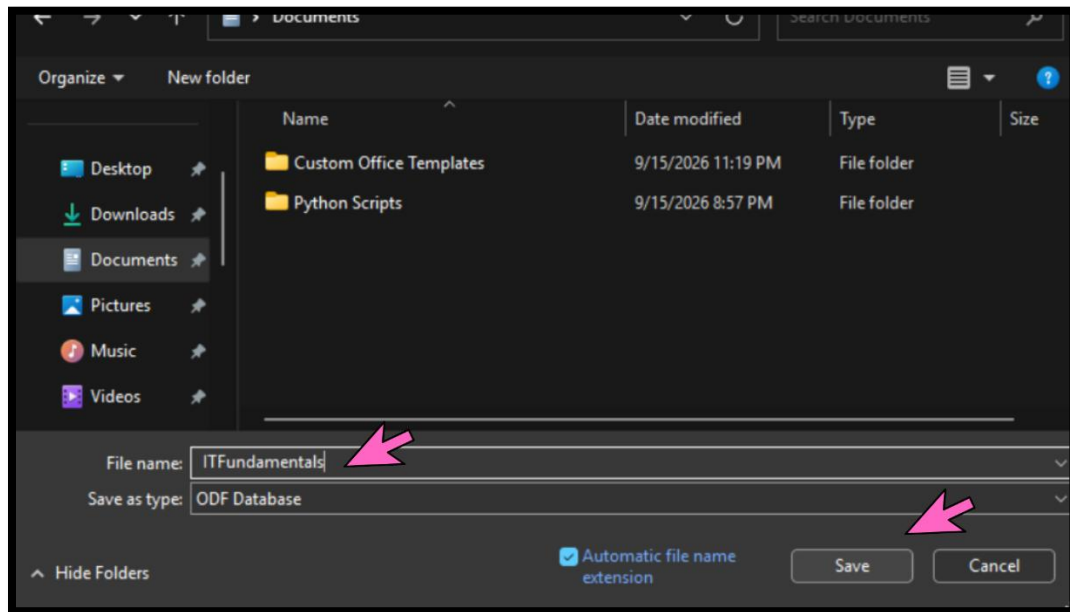


12. In the Save As window, in the navigation pane, click **Documents**.



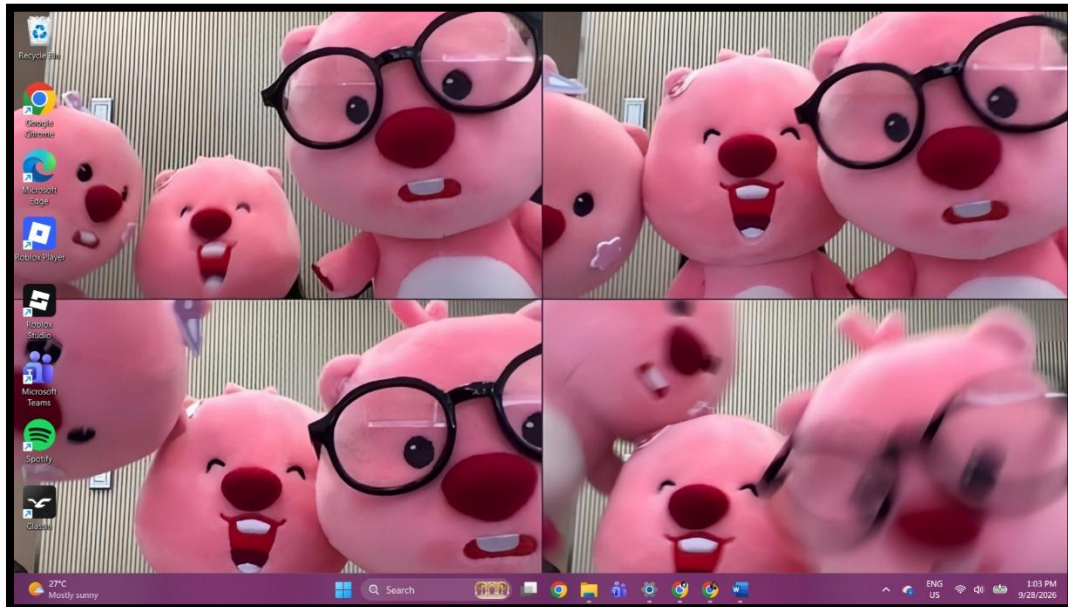


13. In the File name box, type `ITFundamentals` and then click **Save**.



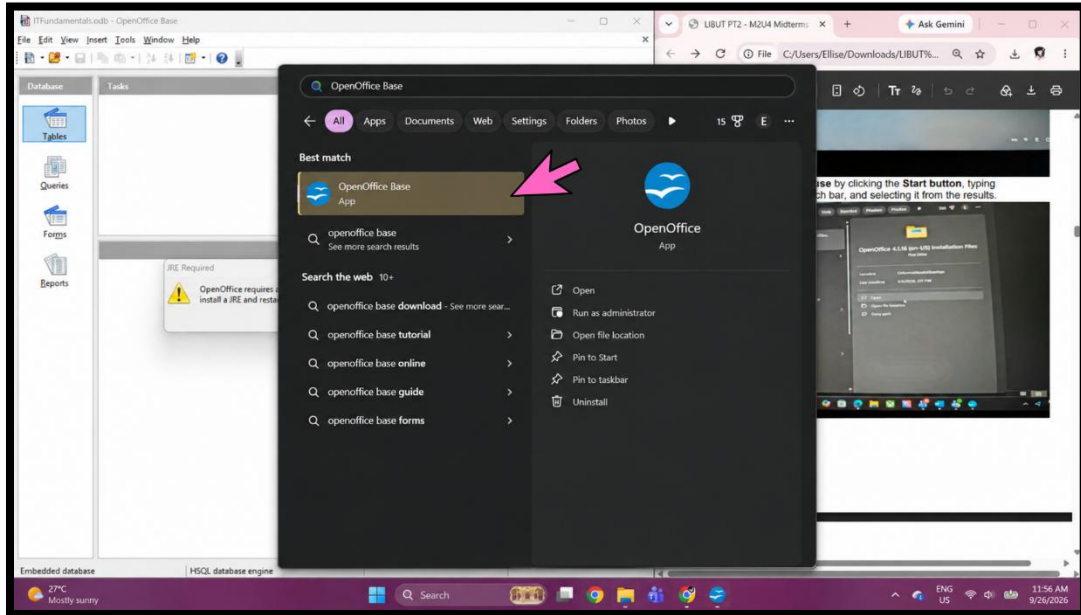
Part 2: Creating a Database

1. If necessary, start your computer and sign in.

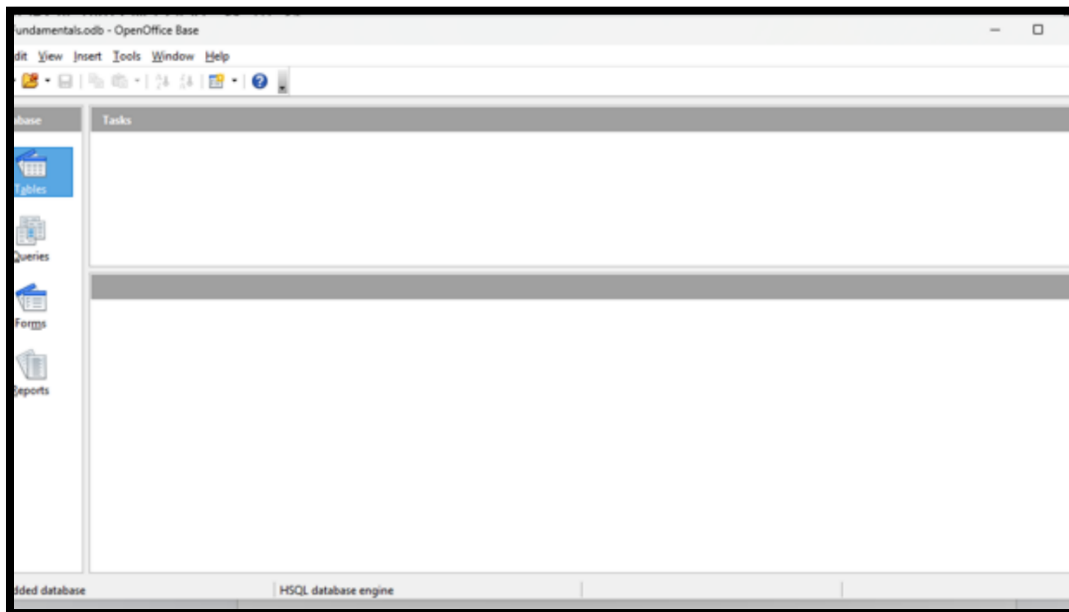




2. Launch **Apache OpenOffice Base** by clicking the **Start button**, typing **OpenOffice Base** in the search bar, and selecting it from the results.



3. In the ITFundamentals - OpenOffice Base window, in the Database Objects pane, ensure **Tables** is selected.



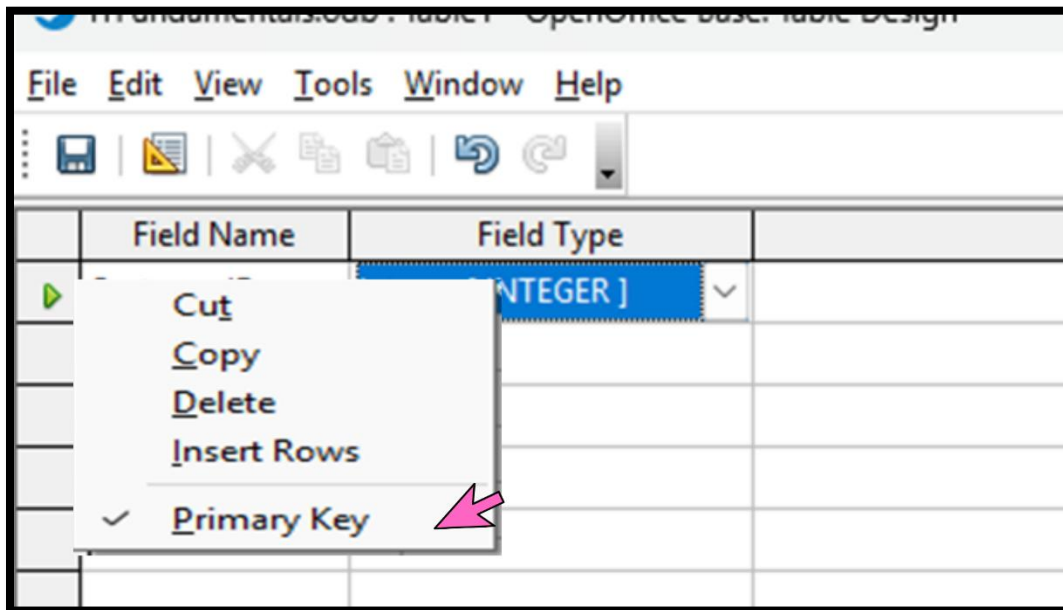


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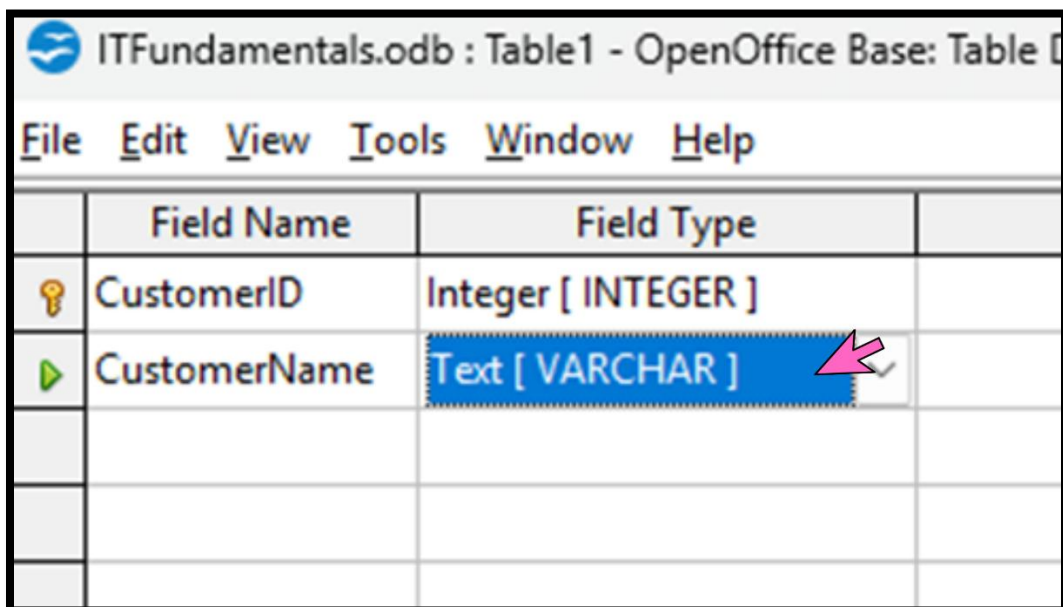
- 
- File Edit View Tools Window Help
- Integer [INTEGER]
- | | Field Name | Field Type |
|---|------------|---------------------|
| ▶ | CustomerID | Integer [INTEGER] |
| | | |
| | | |
| | | |
| | | |
| | | |



6. Right-click the CustomerID row, and then click **Primary Key**.

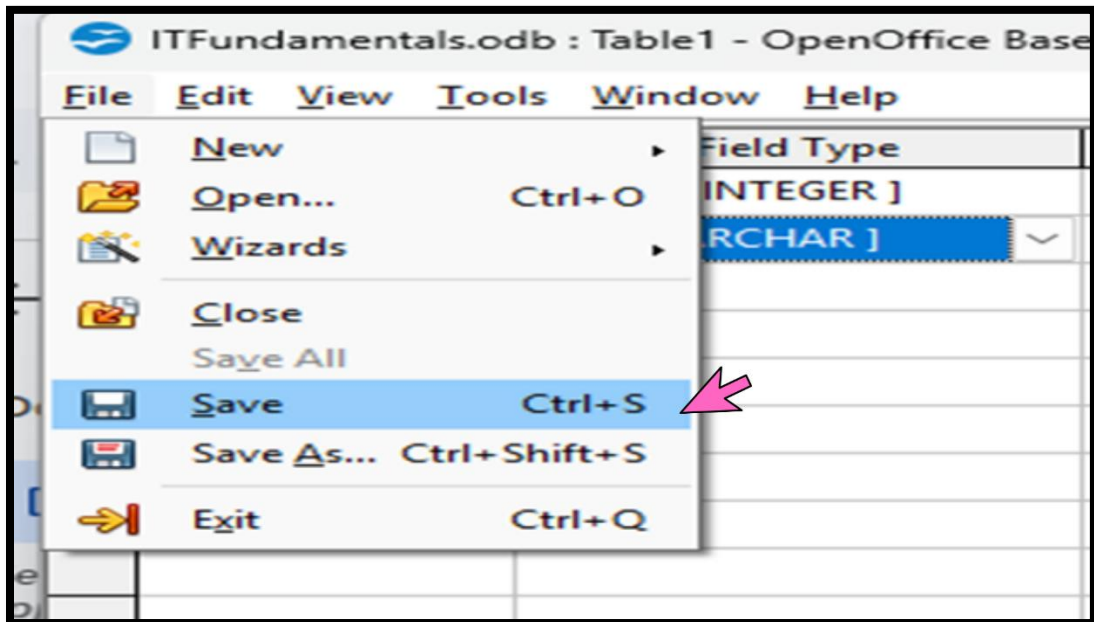


7. In the Field Name column, type CustomerName. In the Field Type column, ensure Text [VARCHAR] is selected.

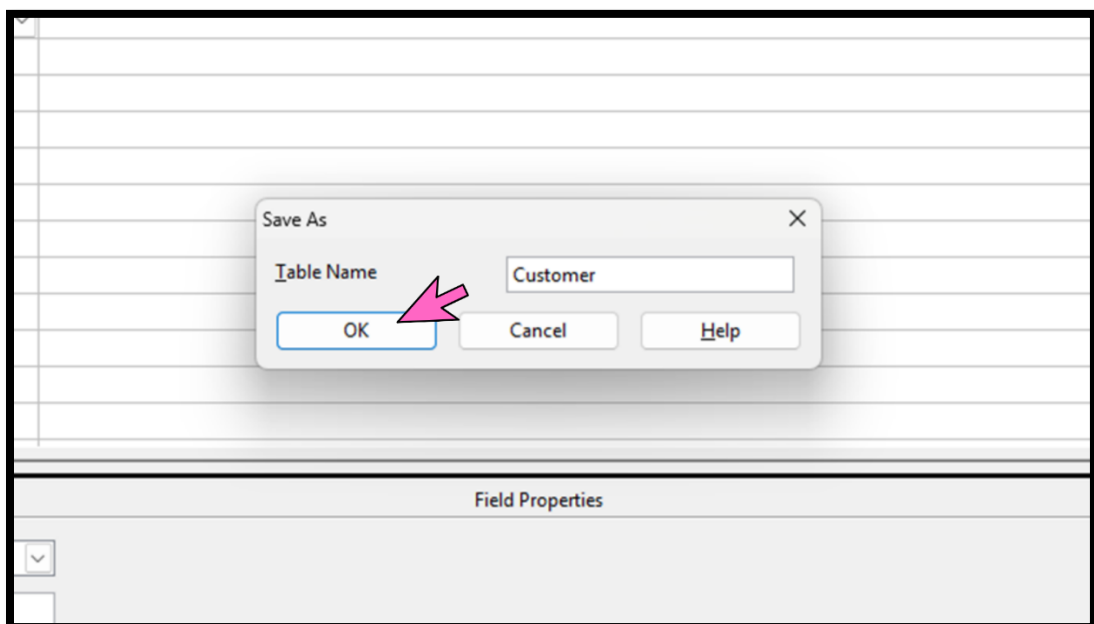




8. From the File menu, click **Save**.

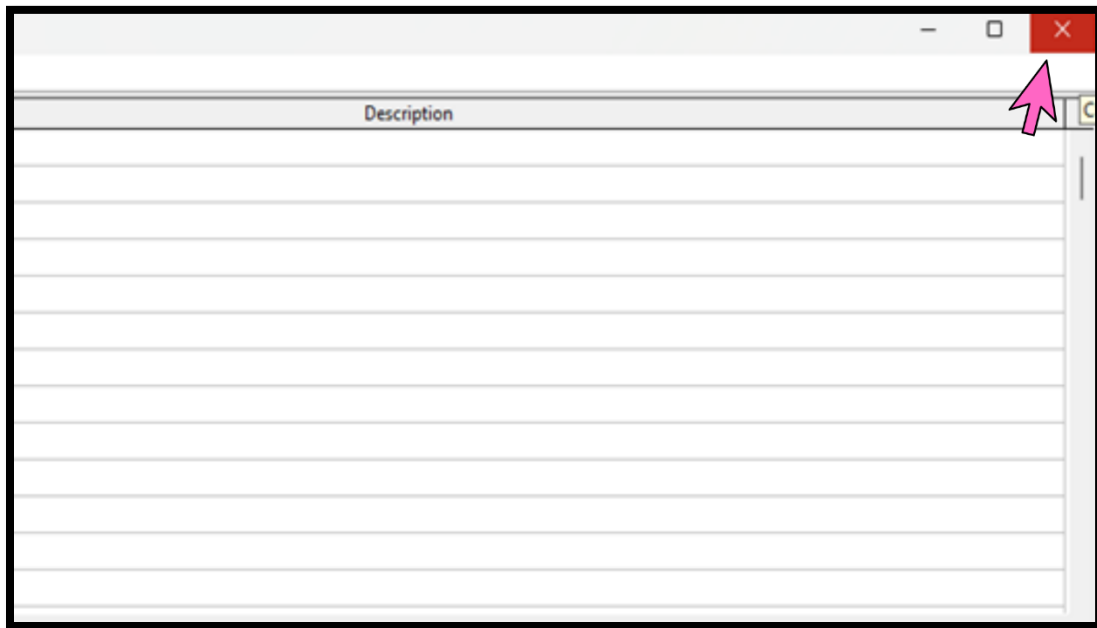


9. In the Save As window, in the Table name box, type `Customer` and then click **OK**.

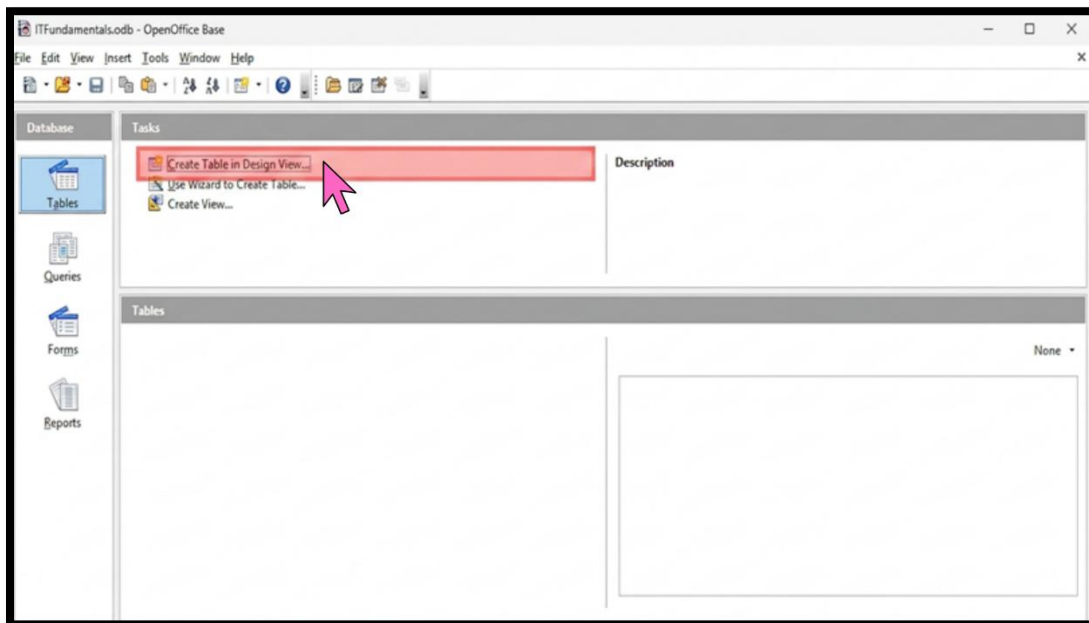




10. Close the Table Design window.



11. In the Tasks pane, click **Create Table in Design View**.





12. In the Table Design window, in the Field Name column, type `ContactID`. In the Field Type column, ensure `Integer [INTEGER]` is selected.

	Field Name	Field Type
	ContactID	Integer [INTEGER]
▶		

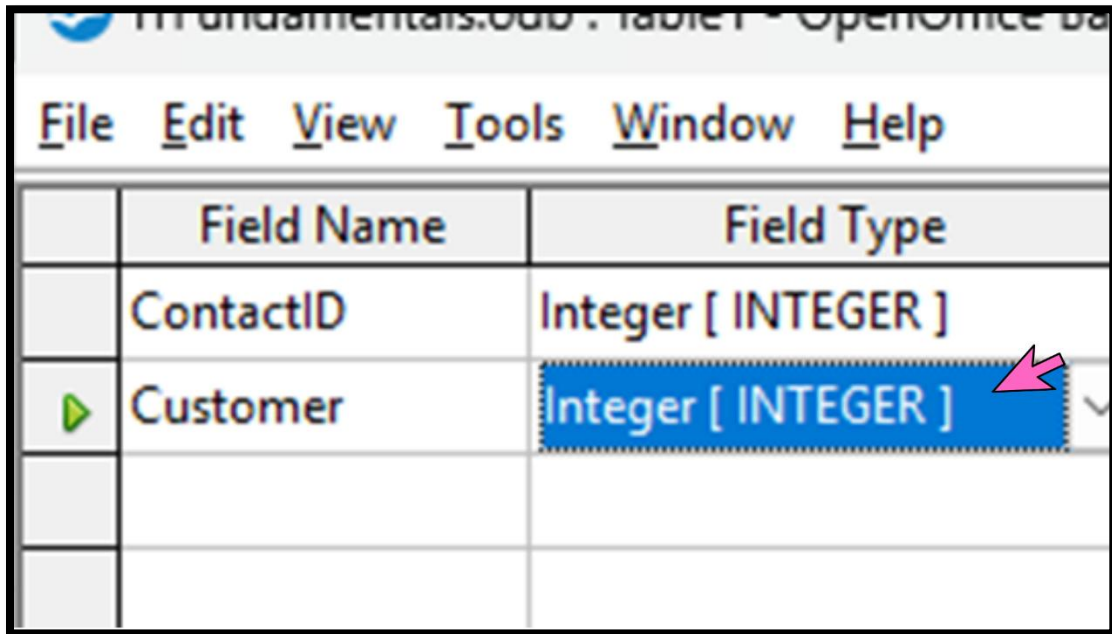
13. Right-click the `ContactID` row, and then click **Primary Key**.

	Field Name	Field Type
	ContactID	Integer [INTEGER]
▶		

- Cut
- Copy
- Delete
- Insert Rows
- ✓ Primary Key

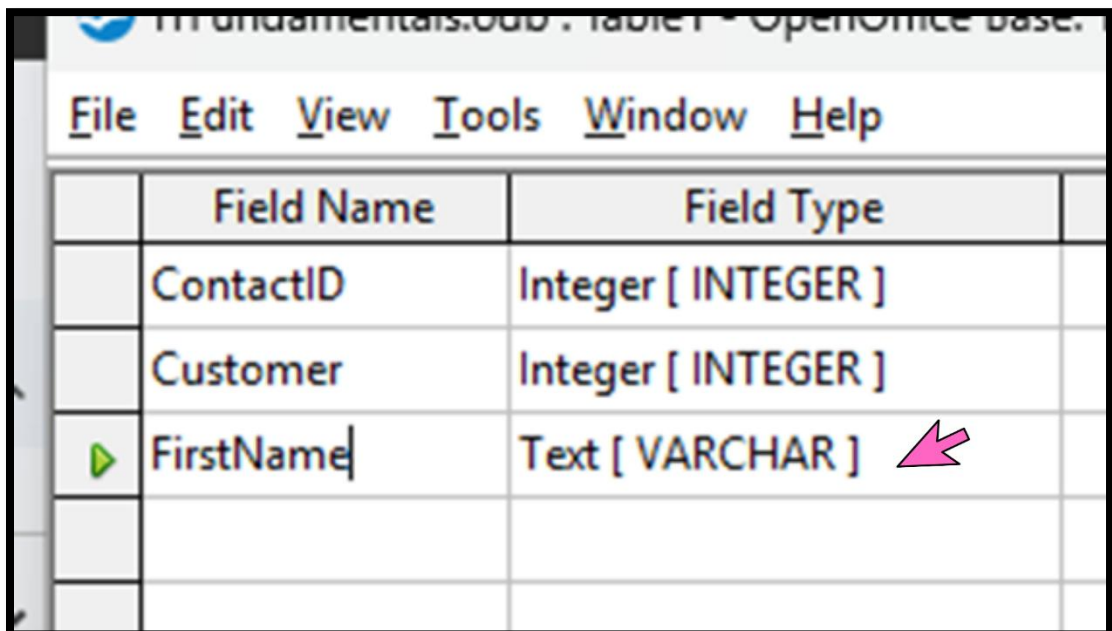


14. In the Field Name column, type `Customer`. In the Field Type column, ensure `Integer [INTEGER]` is selected.



	Field Name	Field Type
	ContactID	Integer [INTEGER]
▶	Customer	Integer [INTEGER]

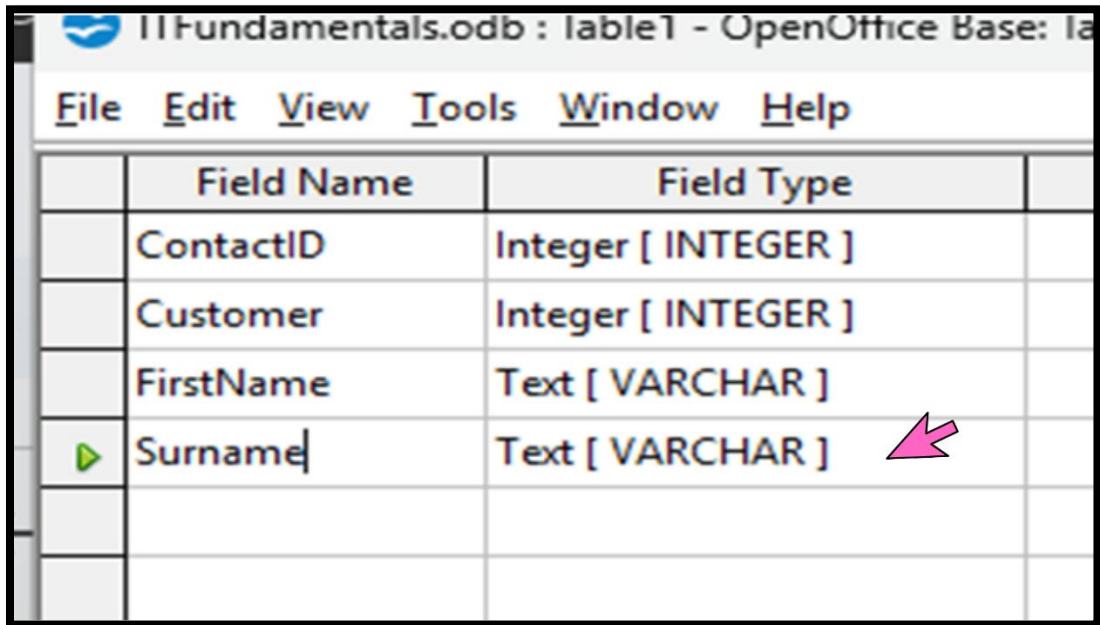
15. In the Field Name column, type `FirstName`. In the Field Type column, ensure `Text [VARCHAR]` is selected.



	Field Name	Field Type	
	ContactID	Integer [INTEGER]	
	Customer	Integer [INTEGER]	
▶	FirstName	Text [VARCHAR]	

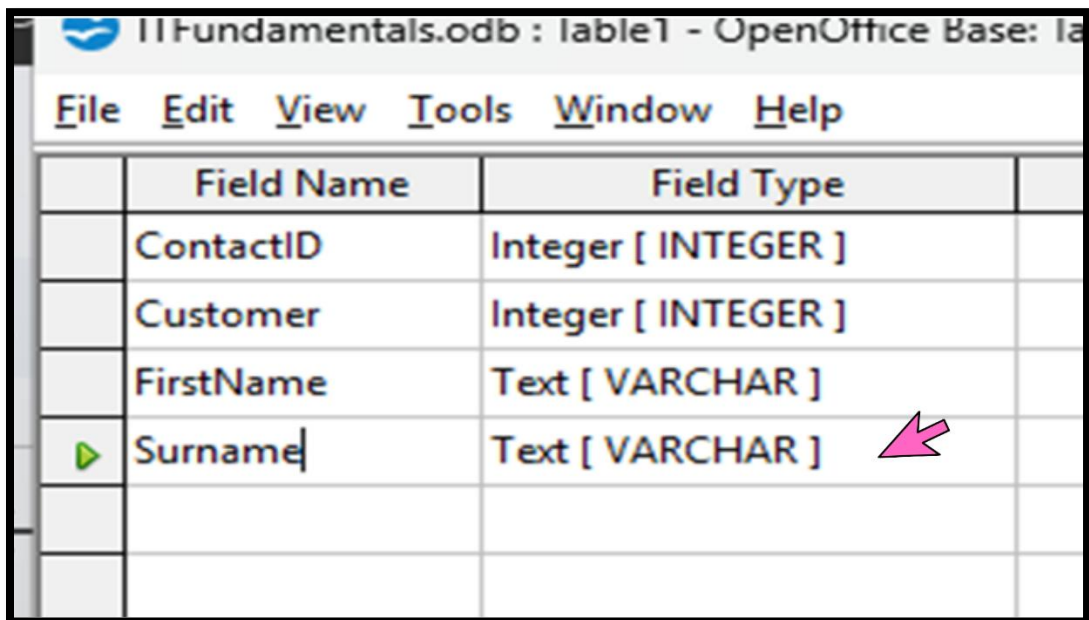


16. In the Field Name column, type Surname. In the Field Type column, ensure Text [VARCHAR] is selected.



	Field Name	Field Type	
	ContactID	Integer [INTEGER]	
	Customer	Integer [INTEGER]	
	FirstName	Text [VARCHAR]	
▶	Surname	Text [VARCHAR]	

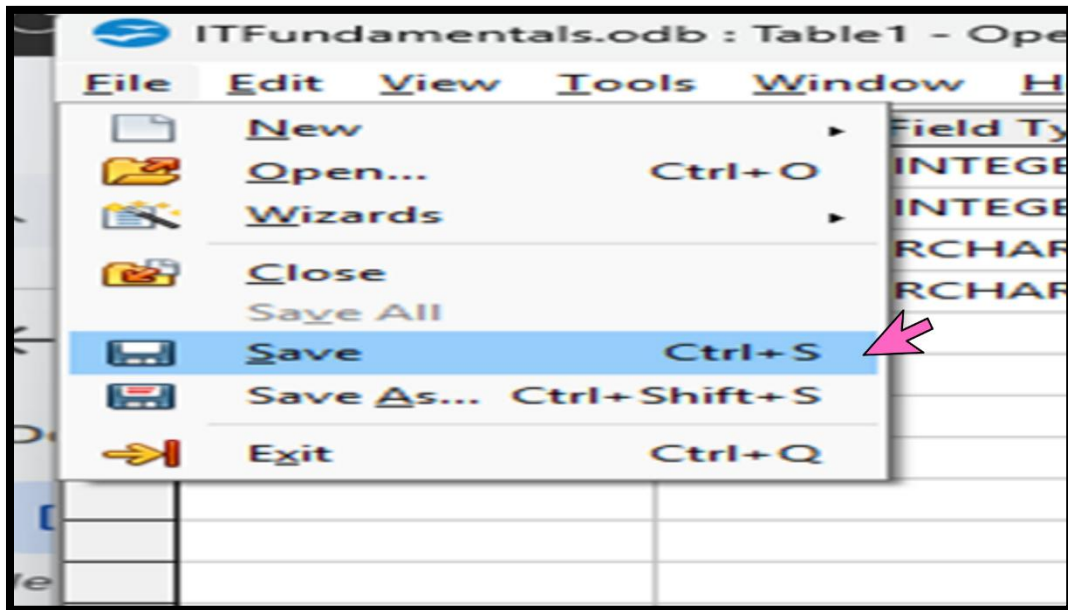
17. In the Field Name column, type ContactNum. In the Field Type column, ensure Text [VARCHAR] is selected.



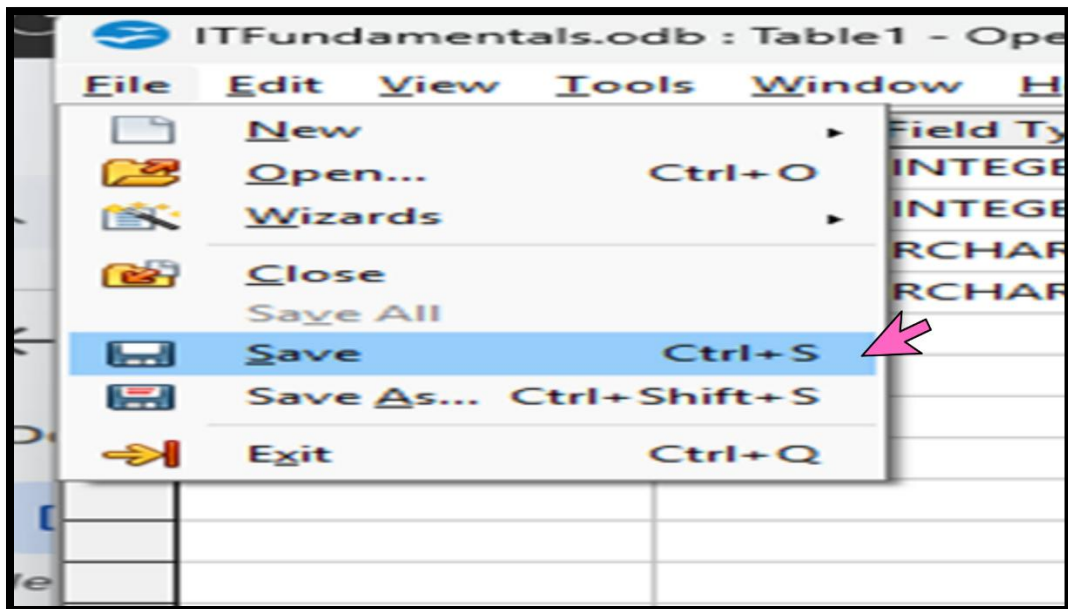
	Field Name	Field Type	
	ContactID	Integer [INTEGER]	
	Customer	Integer [INTEGER]	
	FirstName	Text [VARCHAR]	
▶	Surname	Text [VARCHAR]	



18. From the File menu, click **Save**.

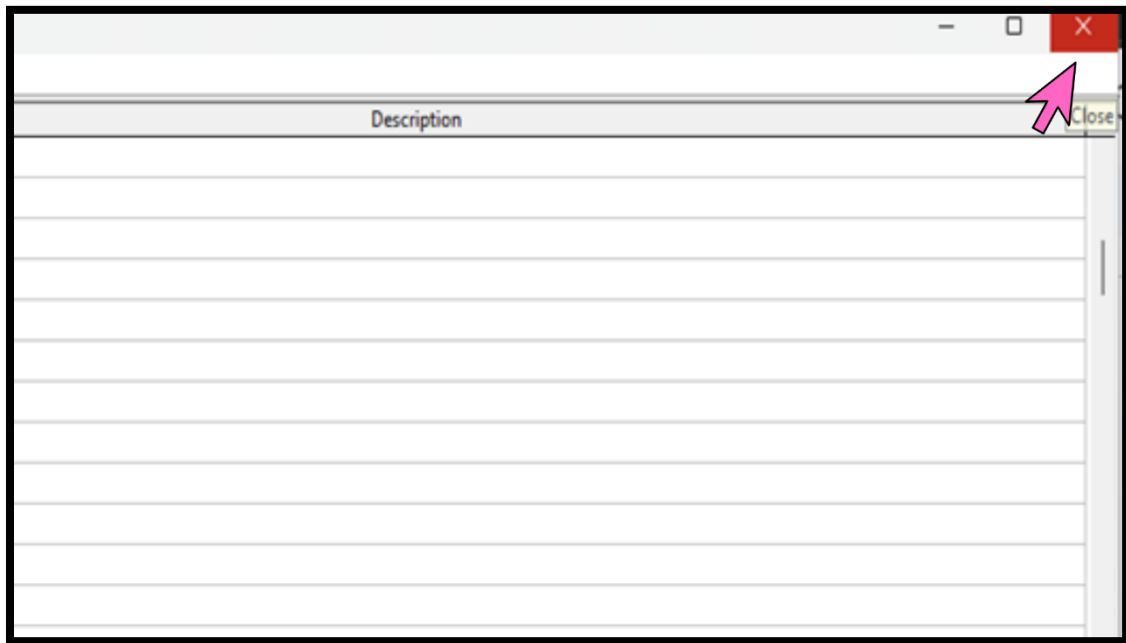


19. In the Save As window, in the Table name box, type **Contact** and then click **OK**.



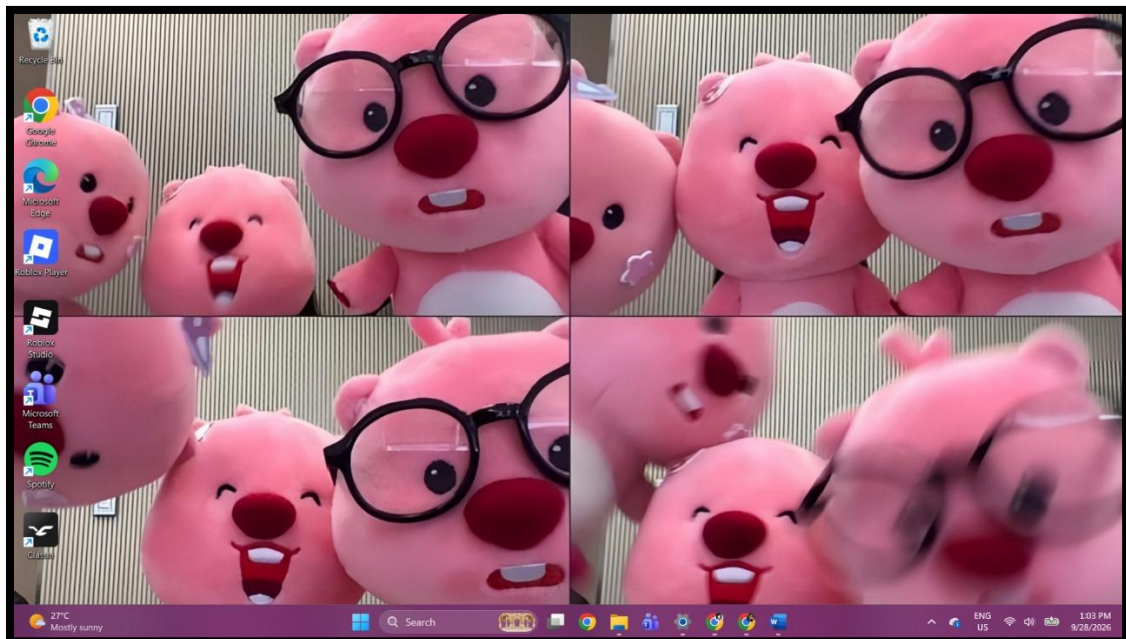


20. Close the Table Design window.



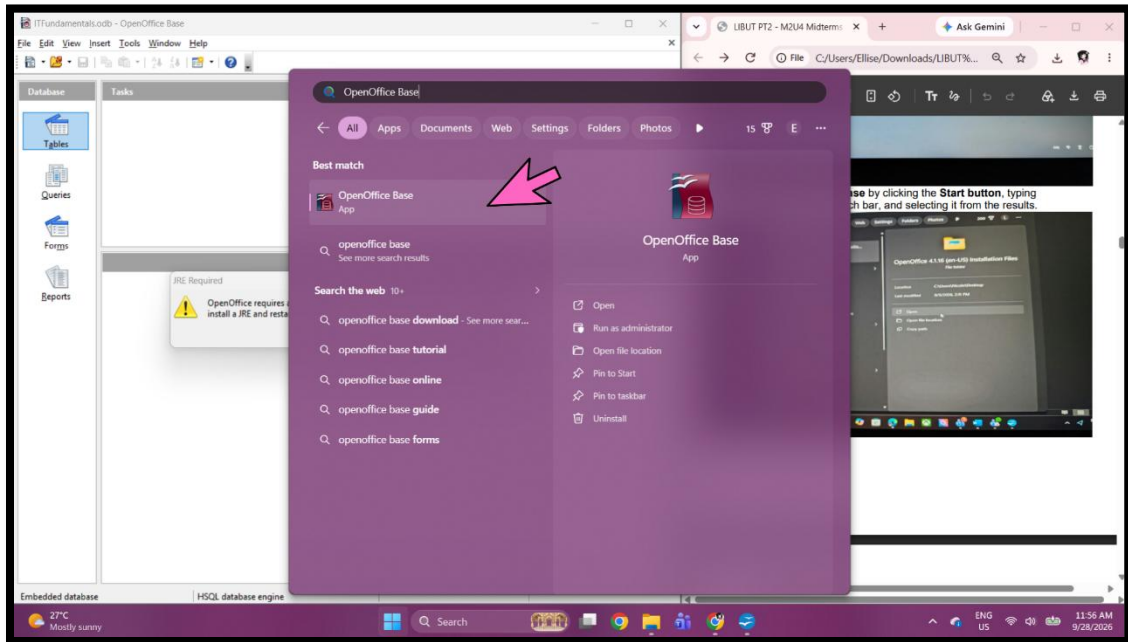
Part 3: Creating a Relationship

1. If necessary, start your computer and sign in.

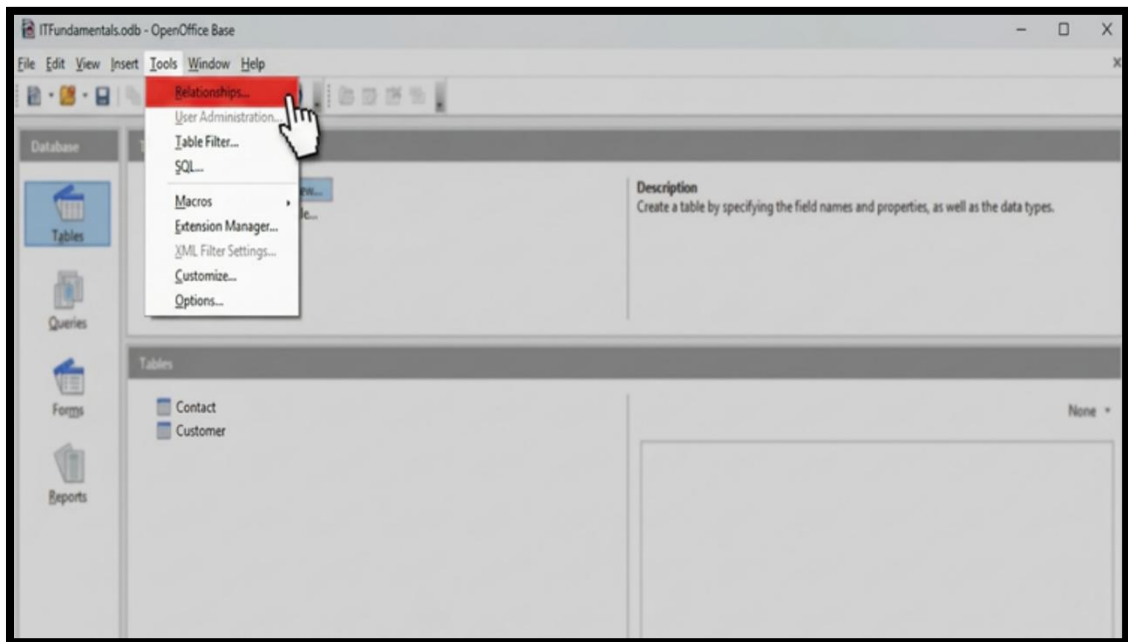




2. Launch **Apache OpenOffice Base** by clicking the **Start button**, typing **OpenOffice Base** in the search bar, and selecting it from the results.

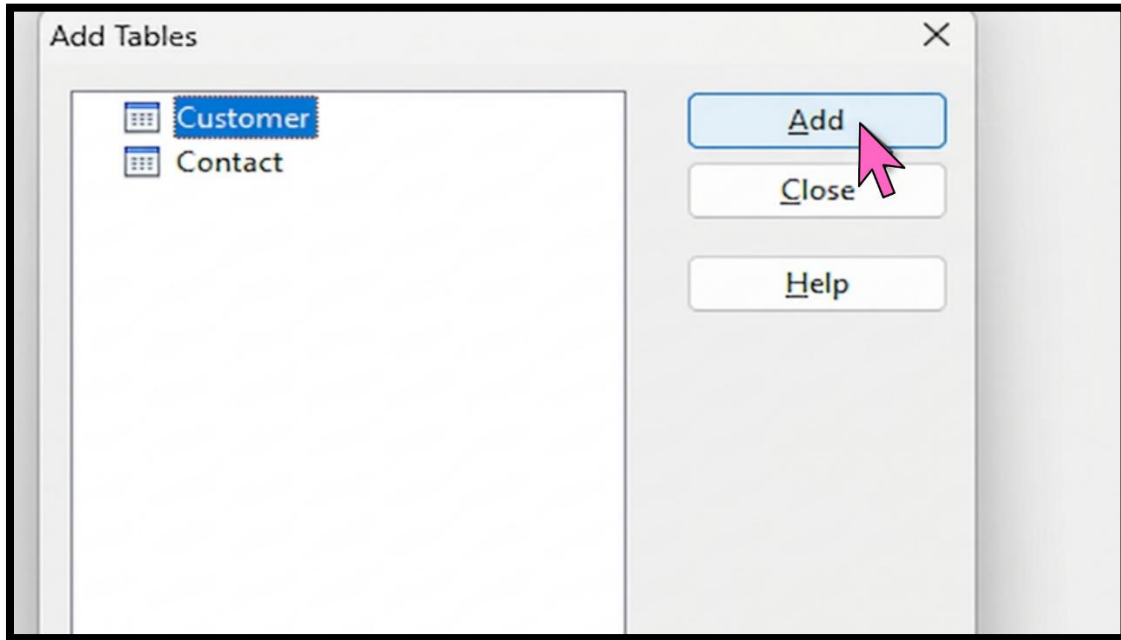


3. From the **Tools** menu, click **Relationships**.

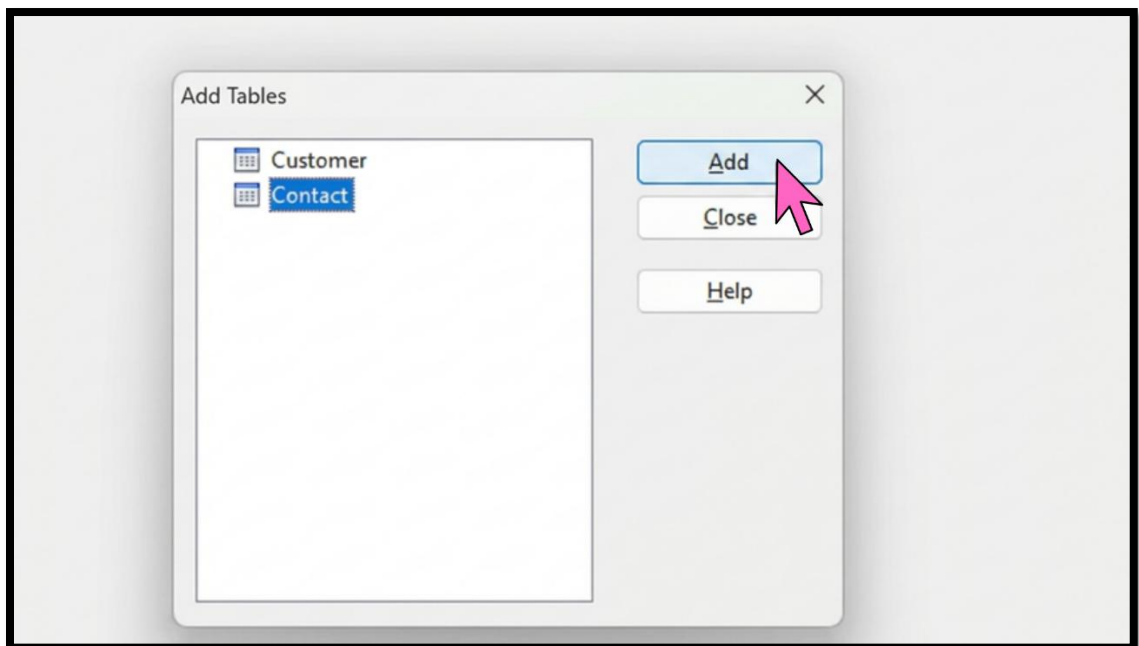




4. In the Relationships window, in the Add Tables window, click **Customer**, and then click **Add**.

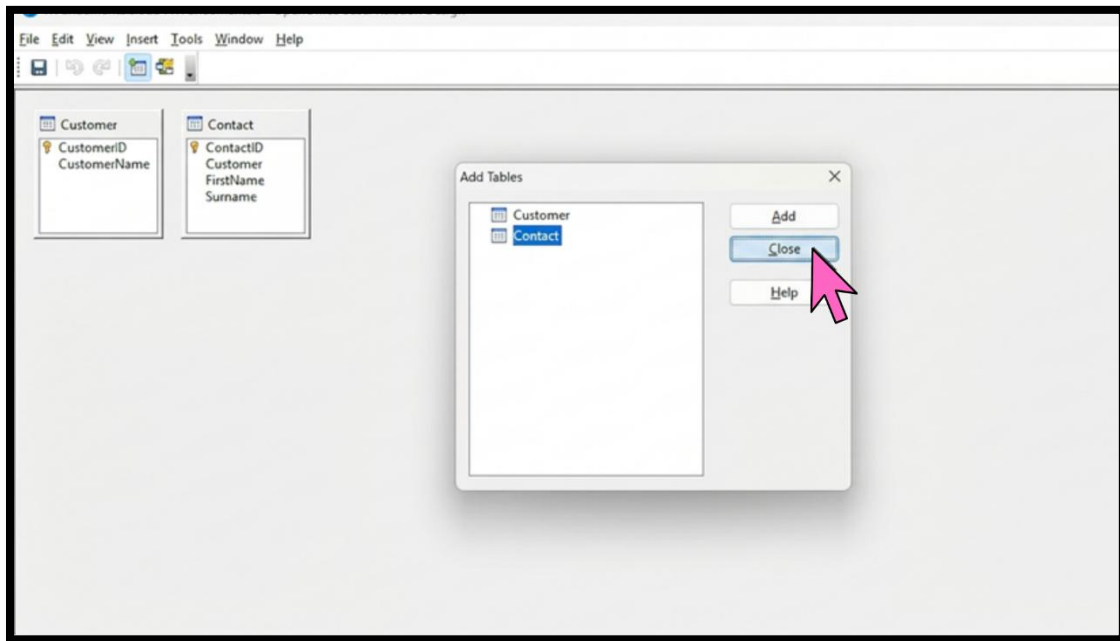


5. Click **Contact**, and then click **Add**.

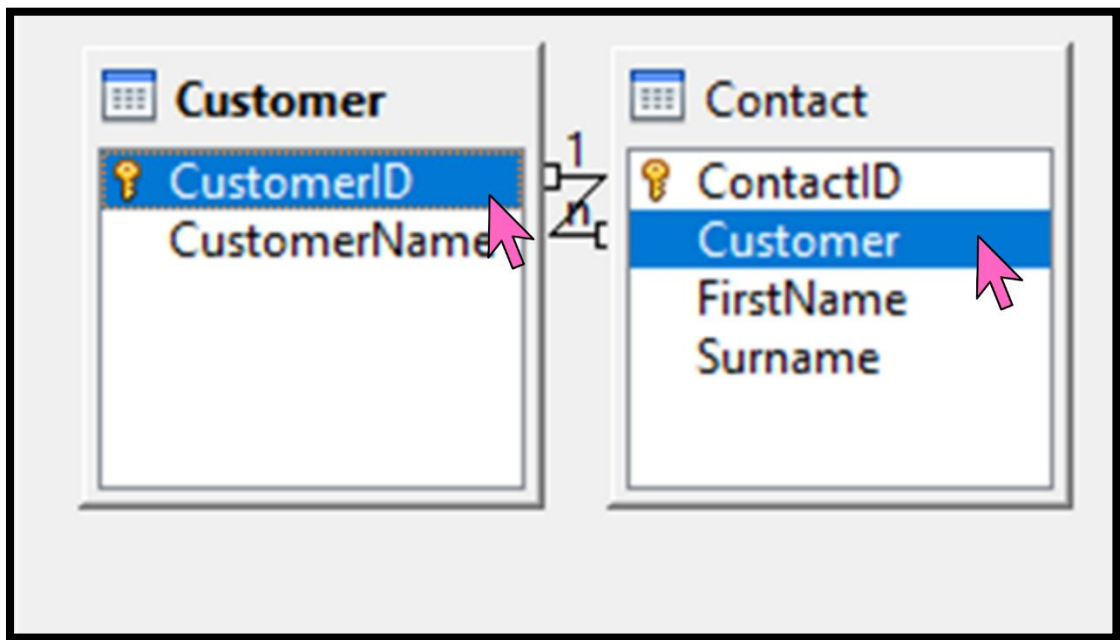




6. Click **Close**.

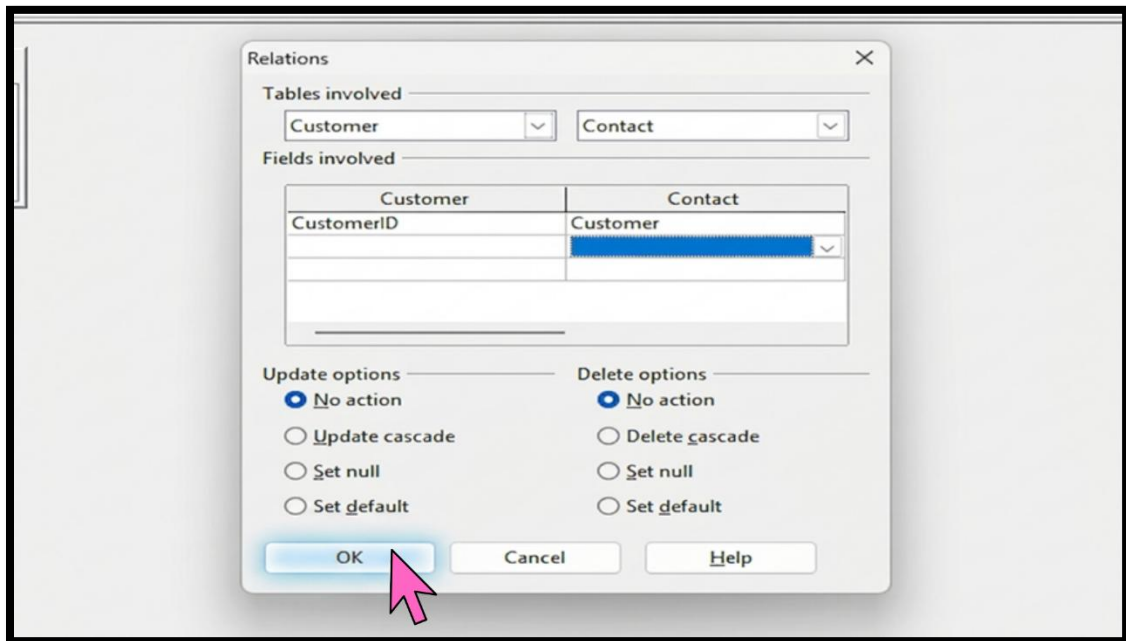


7. In the Relationships window, click and drag the `CustomerID` field from the `Customer` table to the `Customer` field in the `Contact` table.

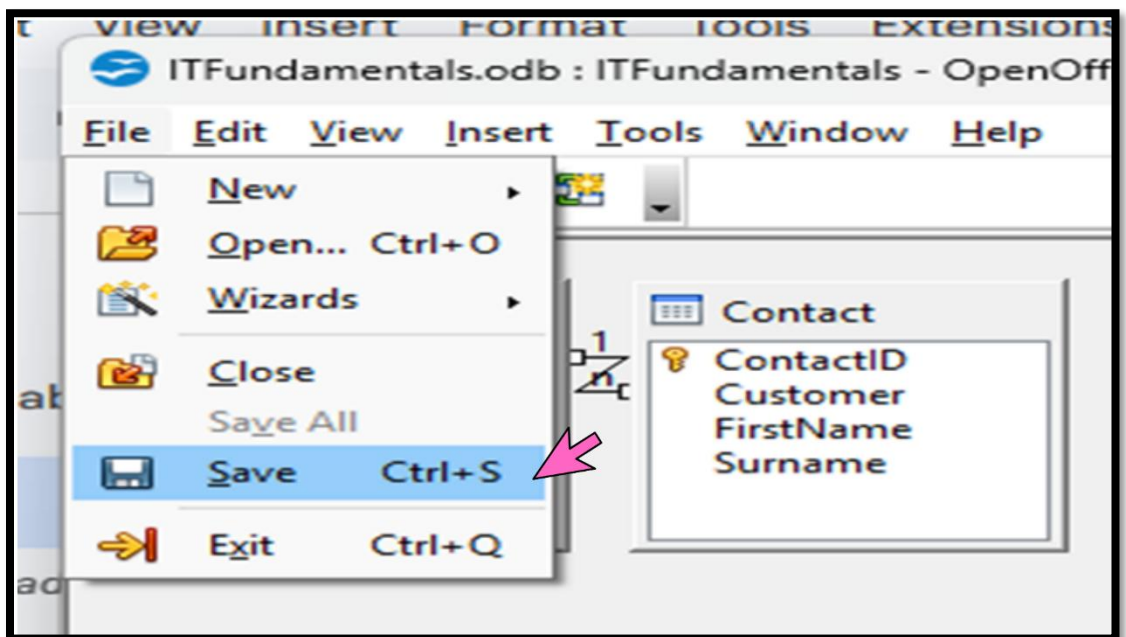




8. In the Relate fields window, click **OK**.



9. From the File menu, click **Save**.



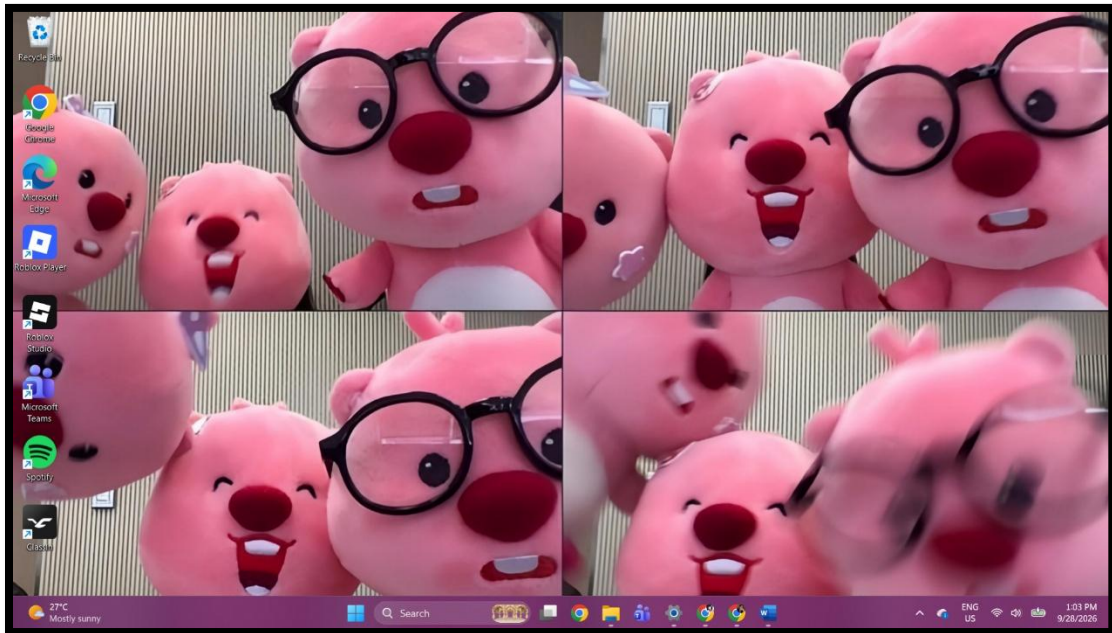


10. Close the Relationships window.



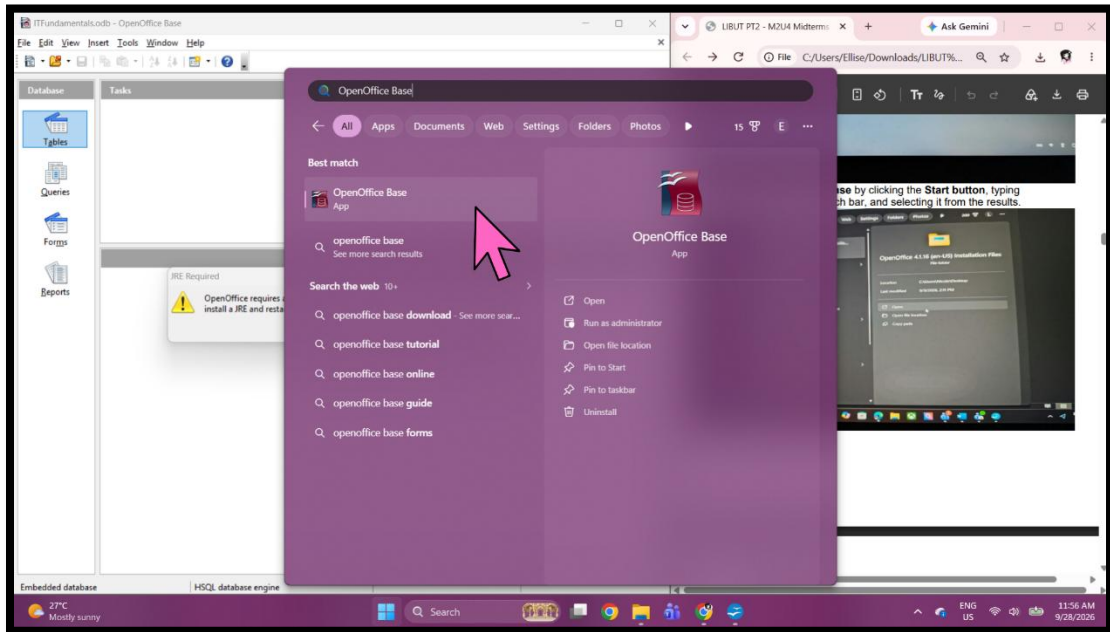
Part 4: Creating a Form

1. If necessary, start your computer and sign in.

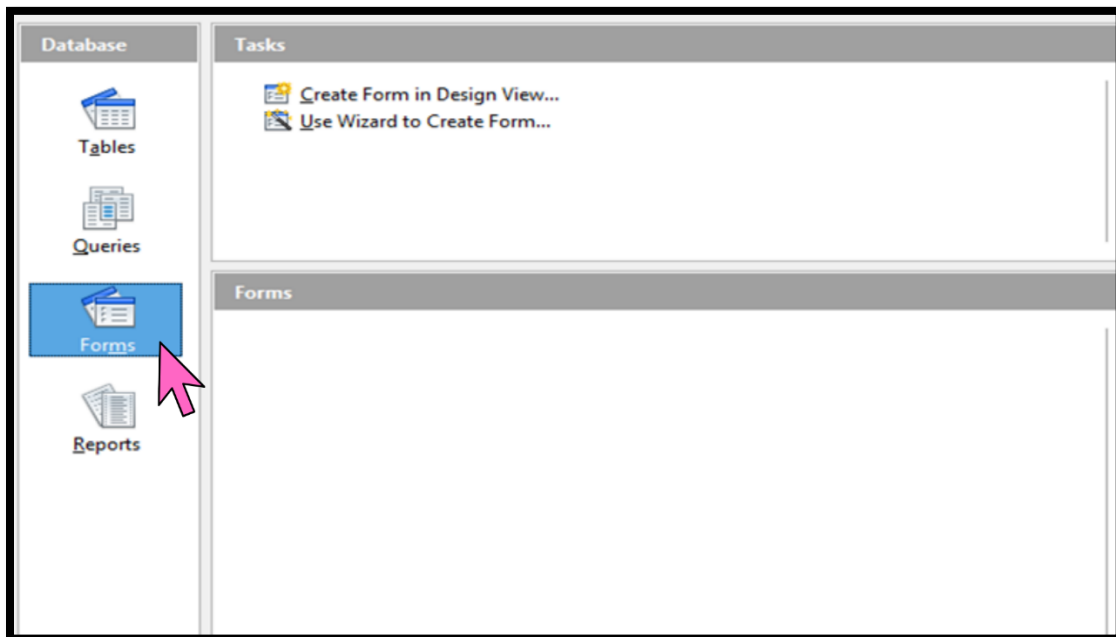




2. Launch **Apache OpenOffice Base** by clicking the **Start button**, typing **OpenOffice Base** in the search bar, and selecting it from the results.

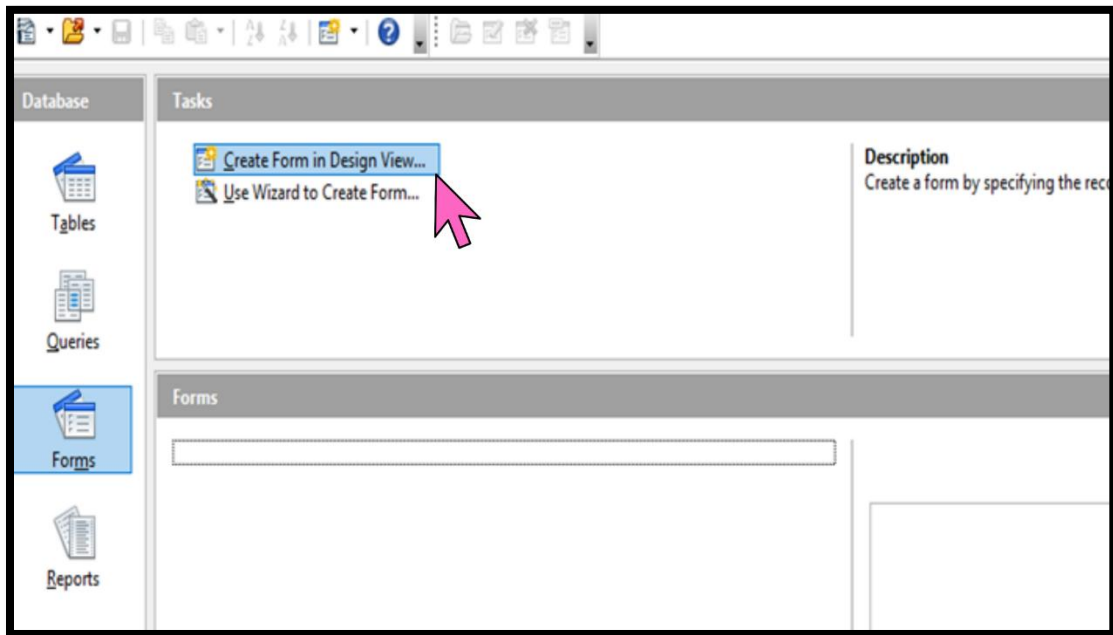


3. In the Database Objects pane, click **Forms**.

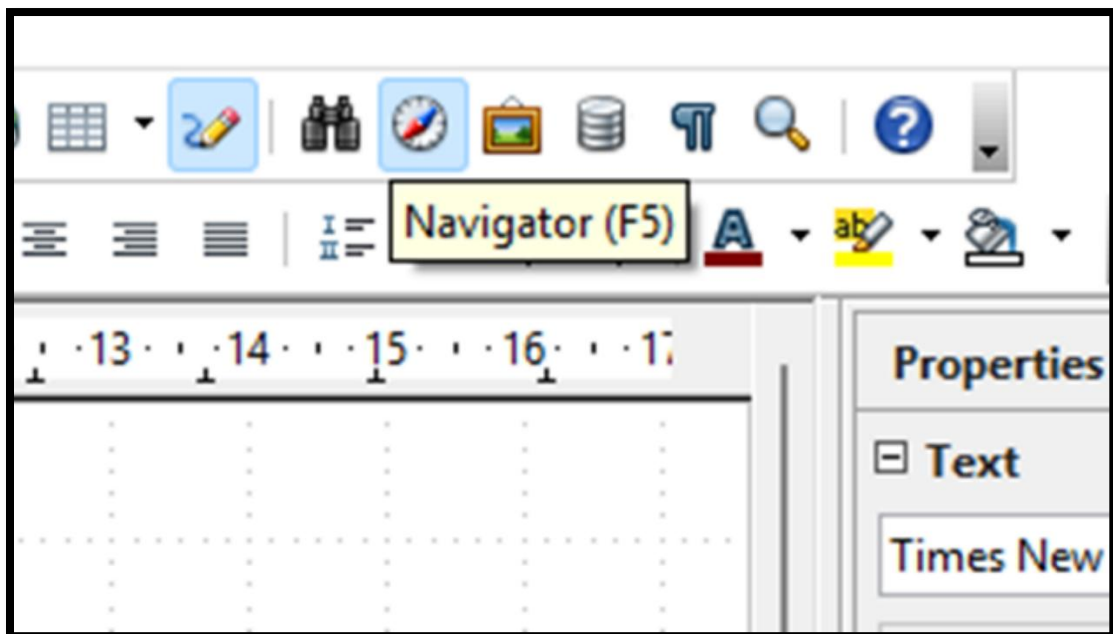




4. In the Tasks pane, click **Create Form in Design View**.

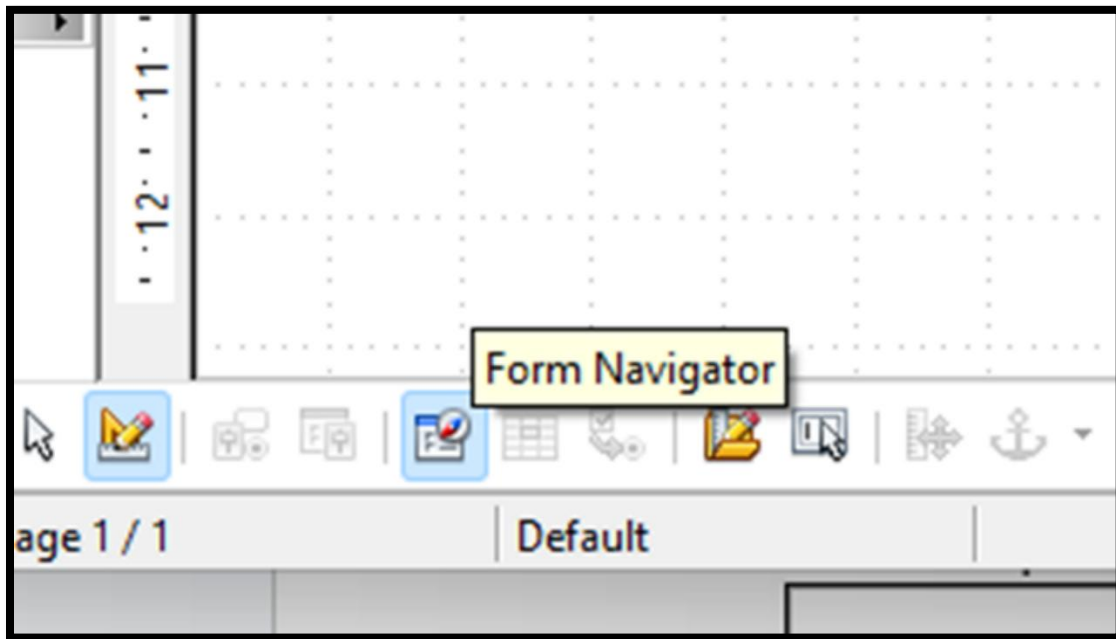


5. In the Form Design window, click the **Form Navigator** button (it looks like a small compass or a document outline icon).

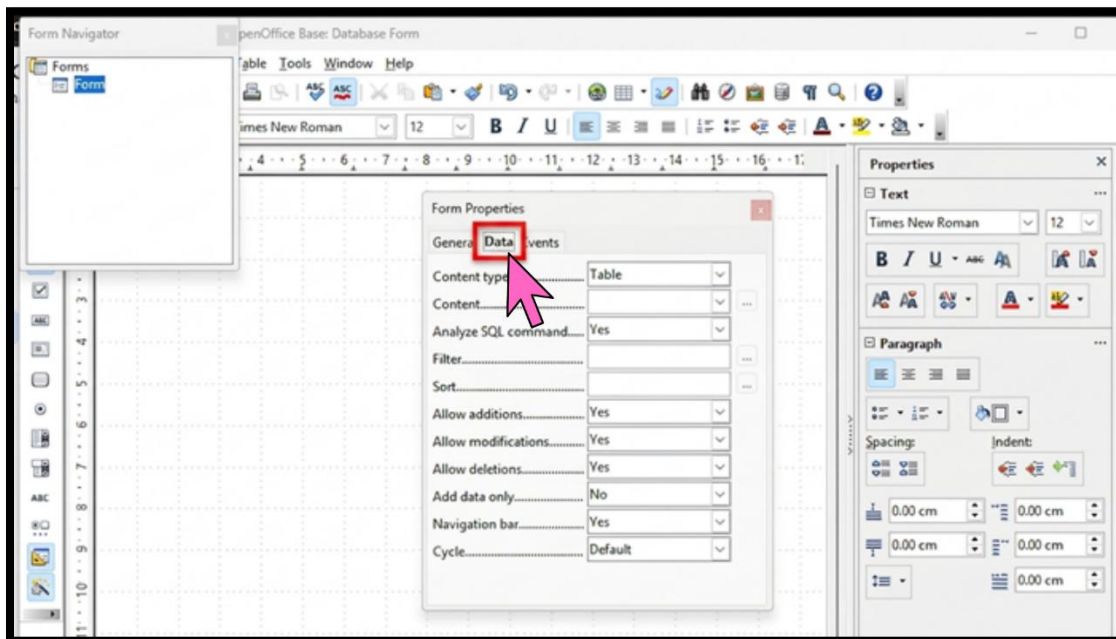




6. In the Form Navigator window, click the **Forms** entry.

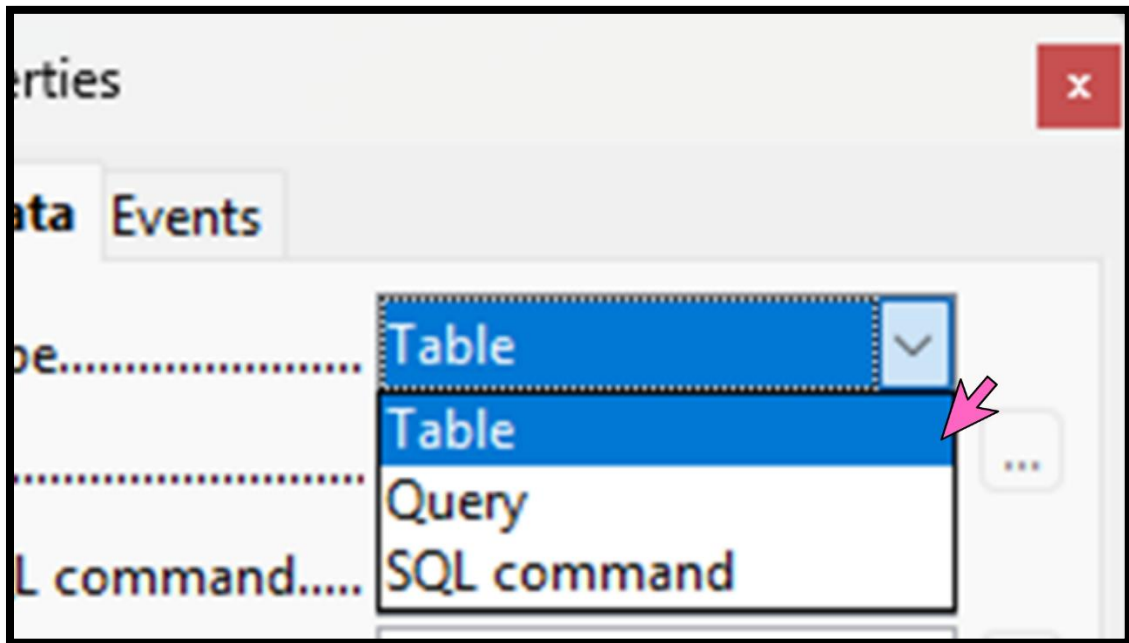


7. In the Properties window, click the **Data** tab.

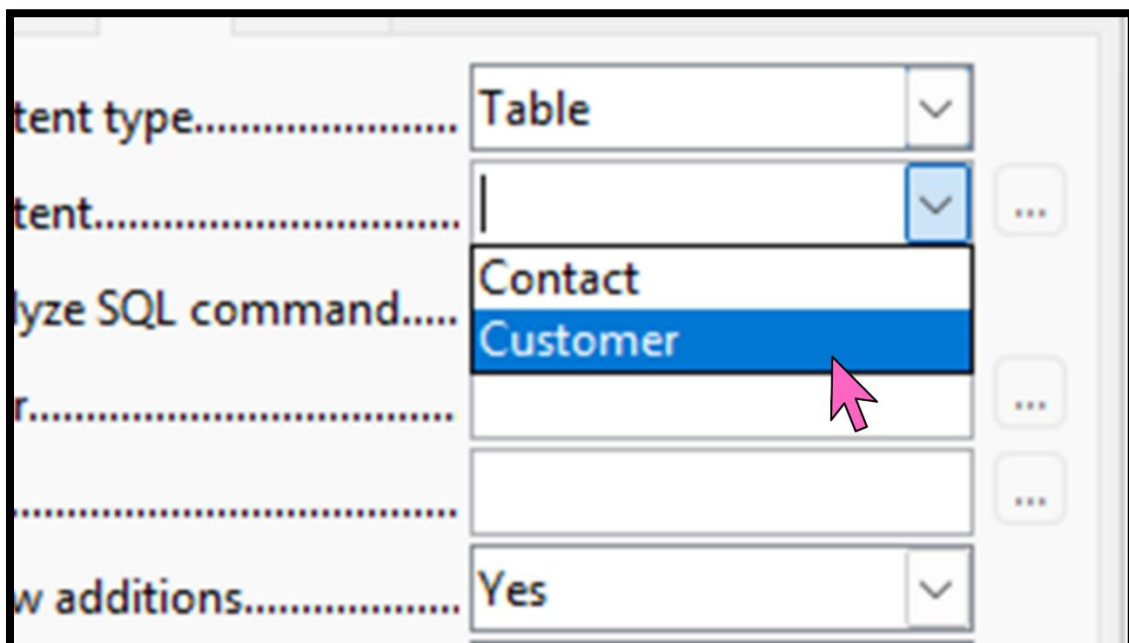




8. In the Data property, in the Content type drop-down list, click **Table**.

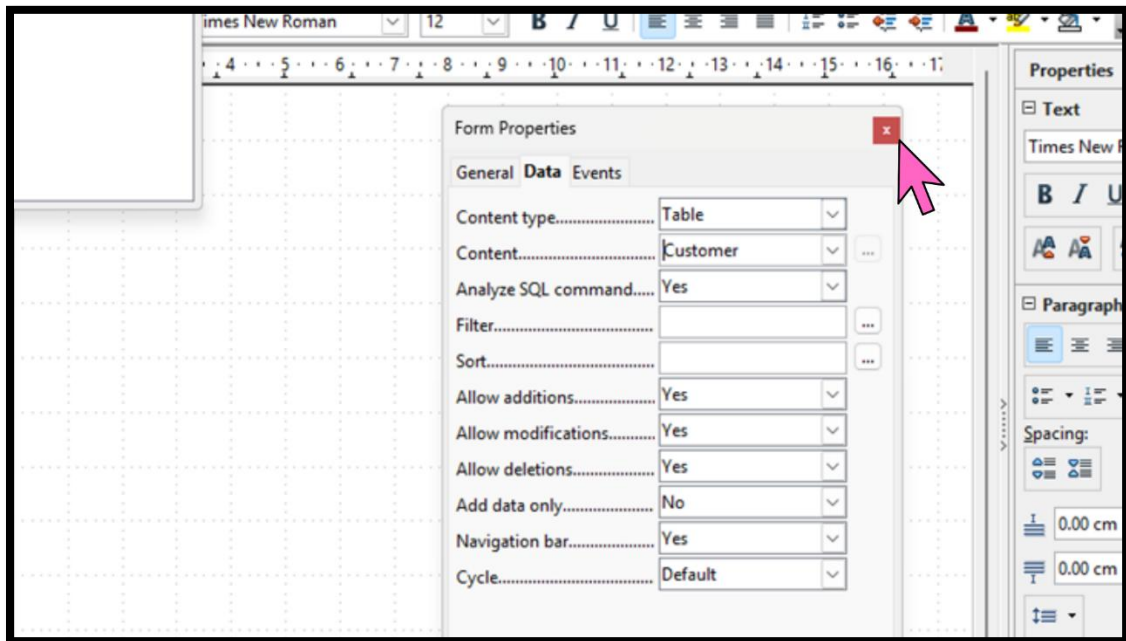


9. In the Content drop-down list, click **Customer**.

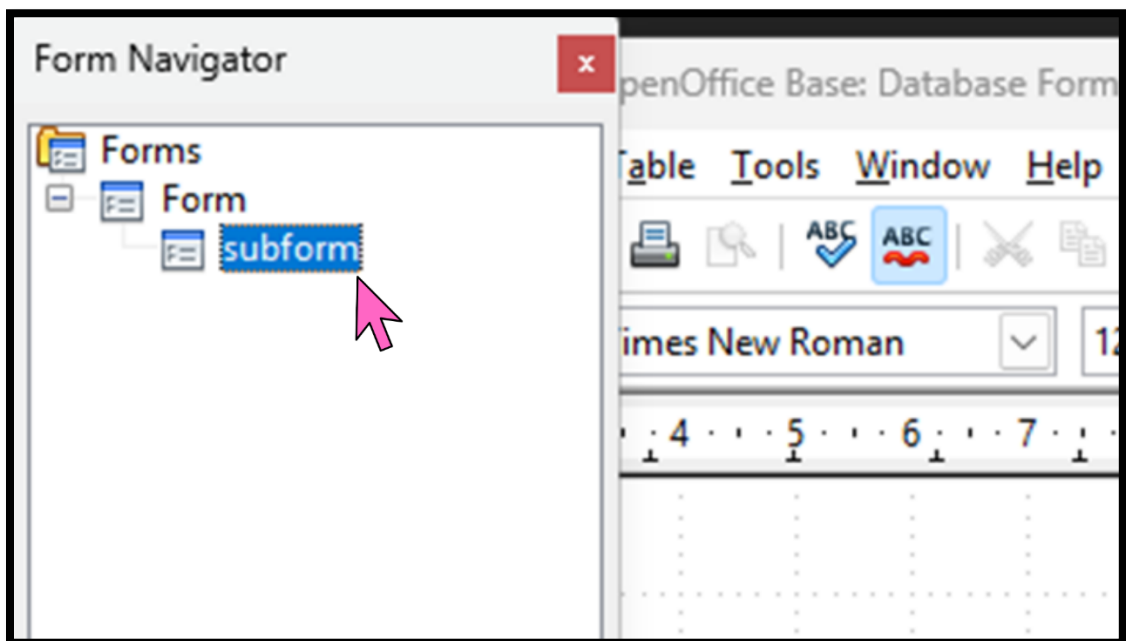




10. Click the **Form Navigator** button again to close the window.

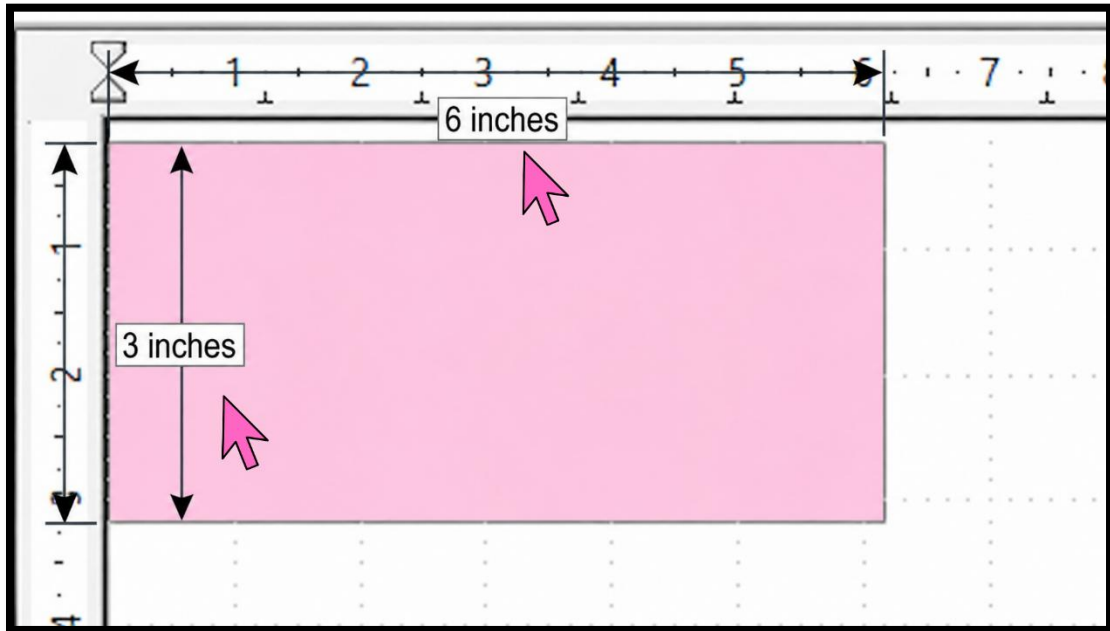


11. From the Insert menu, point to **Form**, and then click **Subform**.

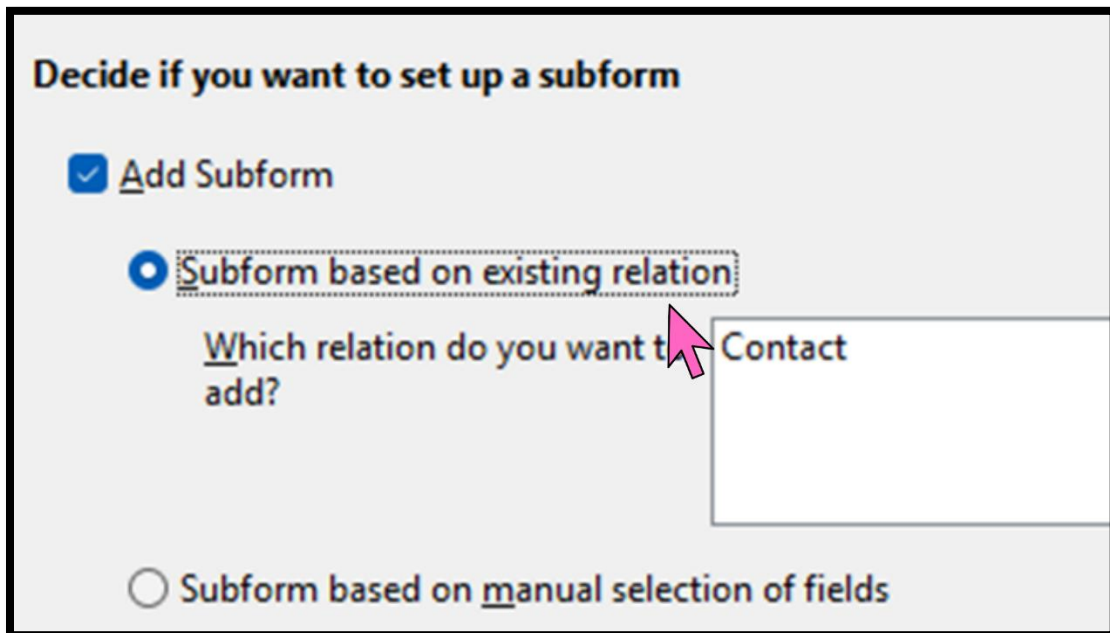




12. In the Form Design window, in the middle of the workspace, draw a rectangle about 6 inches wide and 3 inches high.



13. In the Subform Wizard, ensure **Add Subform Based on Existing Relation** is selected, and then click **Next**.





14. Ensure **Customer** is selected from the top list, and then click **Next**.

Table: Contact

Available fields

- ContactID
- Customer**
- FirstName
- Surname

Fields in the form

> >> < <<

^ v

Binary fields are always listed and selectable from the left list.
If possible, they are interpreted as images.

15. Ensure **Add all fields** is selected (or drag all fields to the right-hand column), and then click **Next**.

Content..... Customer

Analyze SQL command.... Yes

Filter.....

Sort.....

Link master fields.....

Link slave fields..... "CustomerID"

Allow additions..... Yes

Allow modifications..... Yes



16. In the Arrange controls section, ensure **Data Sheet** is selected, and then click **Next**.

Arrangement of the main form Both data sheets are selected

Columnar - Labels Left

STATUS: BOTH DATA SHEETS SELECTED

Arrangement of the subform Both data sheets are selected

Columnar - Labels Left

17. Click **Finish**.

1. Field selection
2. Set up a subform
3. Add subform fields
4. Get joined fields
5. Arrange controls
6. Set data entry
7. Apply styles
8. Set name

Label placement
☒ Align left
☐ Align right

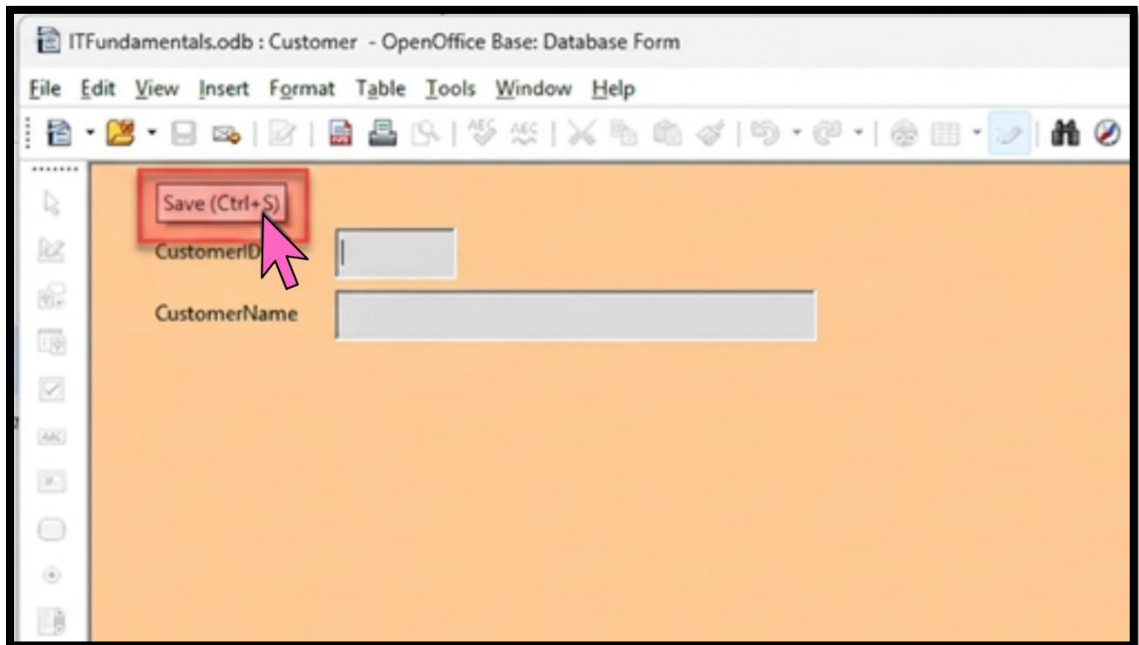
Arrangement of the main form
Columnar - Labels Left

Arrangement of the subform
Columnar - Labels Left

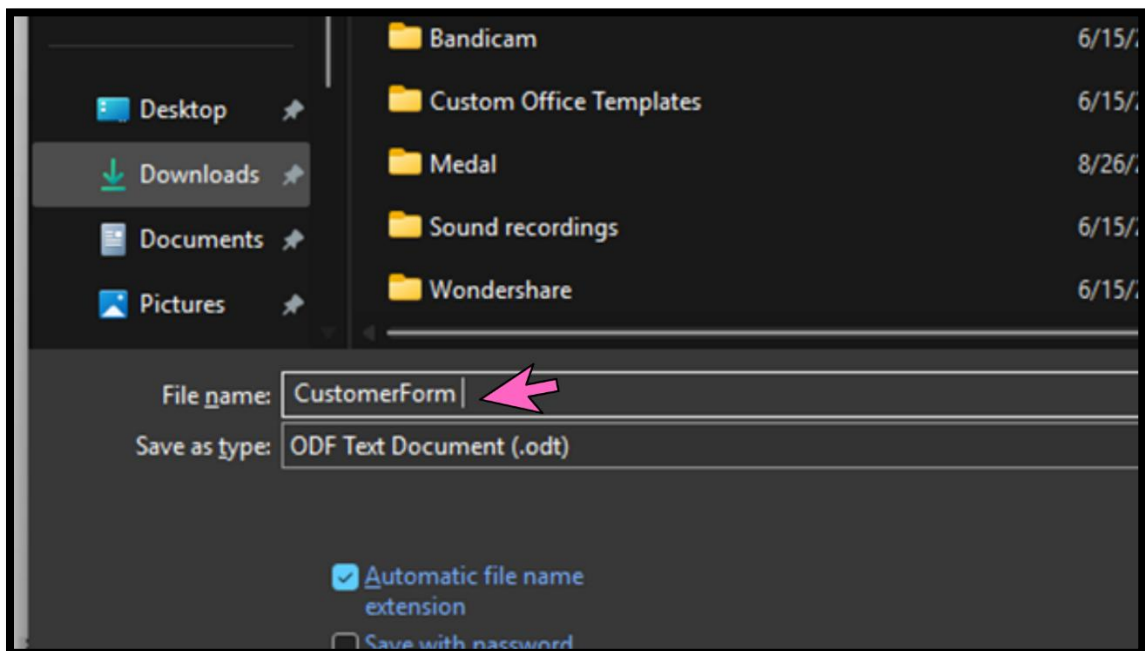
Help < Back Next > Finish Cancel



18. From the Form Design window, click the **Save** button on the toolbar.

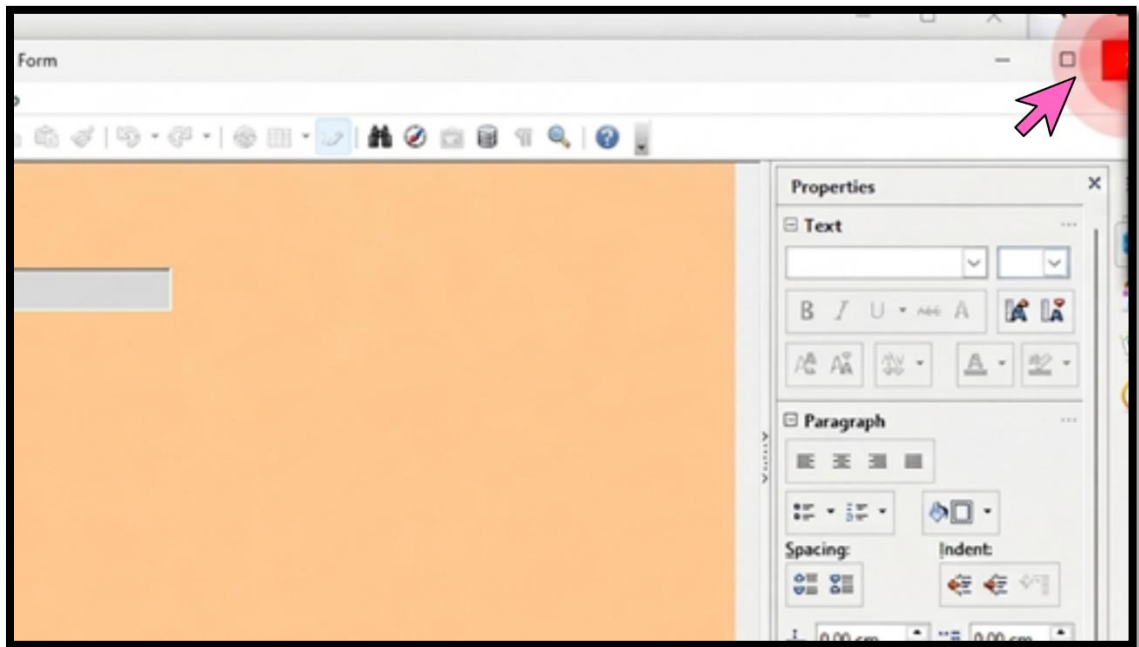


19. In the Save As window, in the Form name box, type CustomerForm and then click **OK**.



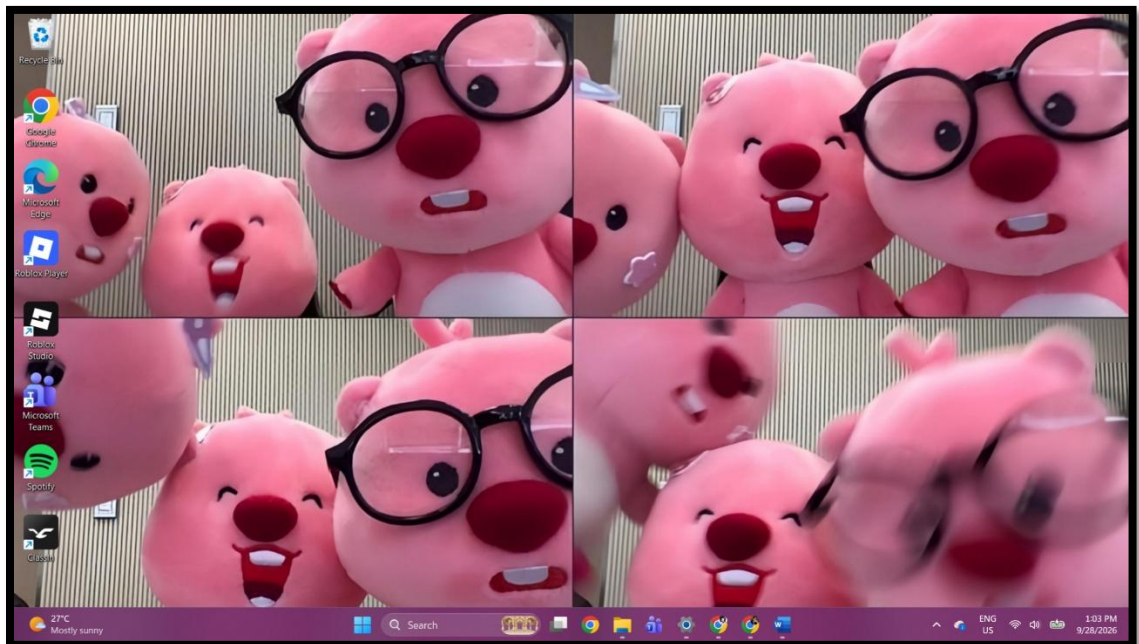


20. Close the Form Design window.



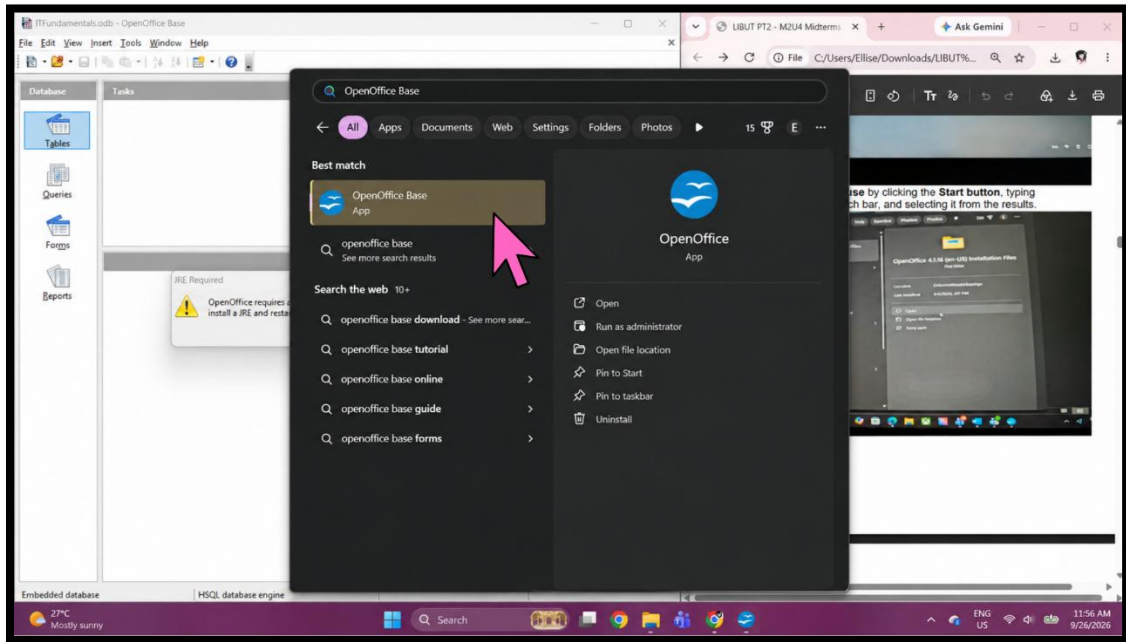
Part 5: Populating the Database

1. If necessary, start your computer and sign in.

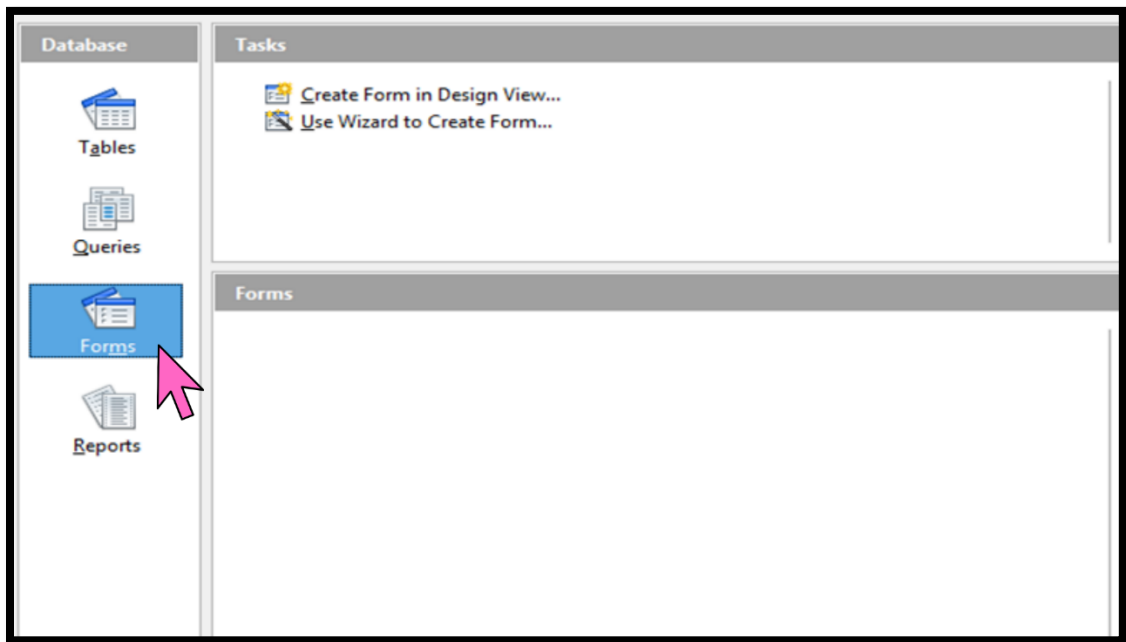




2. Launch **Apache OpenOffice Base** by clicking the **Start button**, typing **OpenOffice Base** in the search bar, and selecting it from the results.

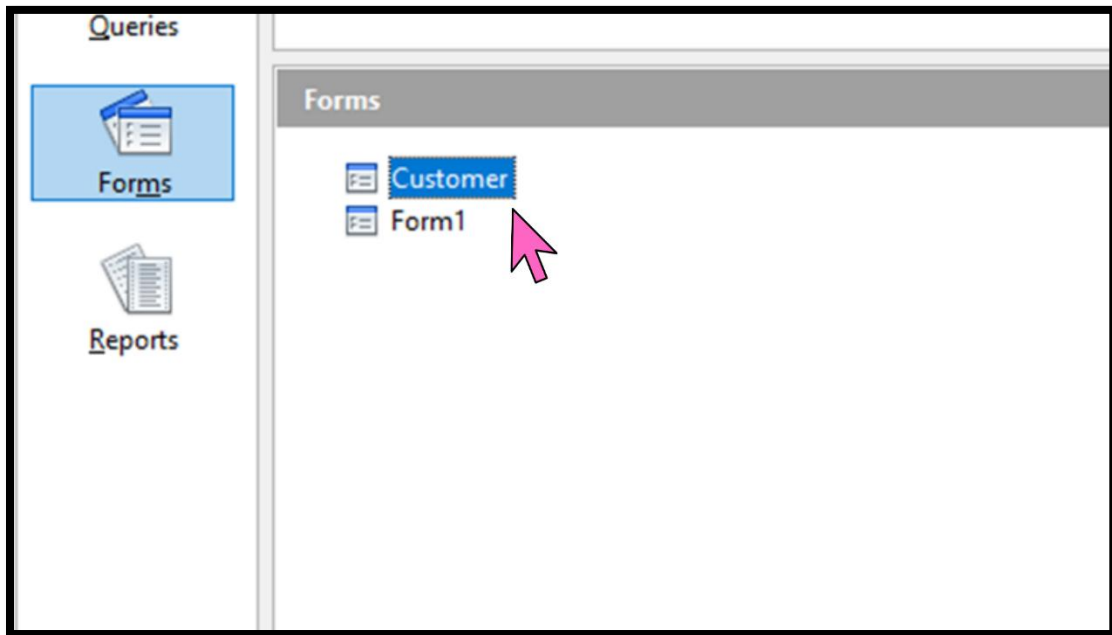


3. In the Database Objects pane, click **Forms**.

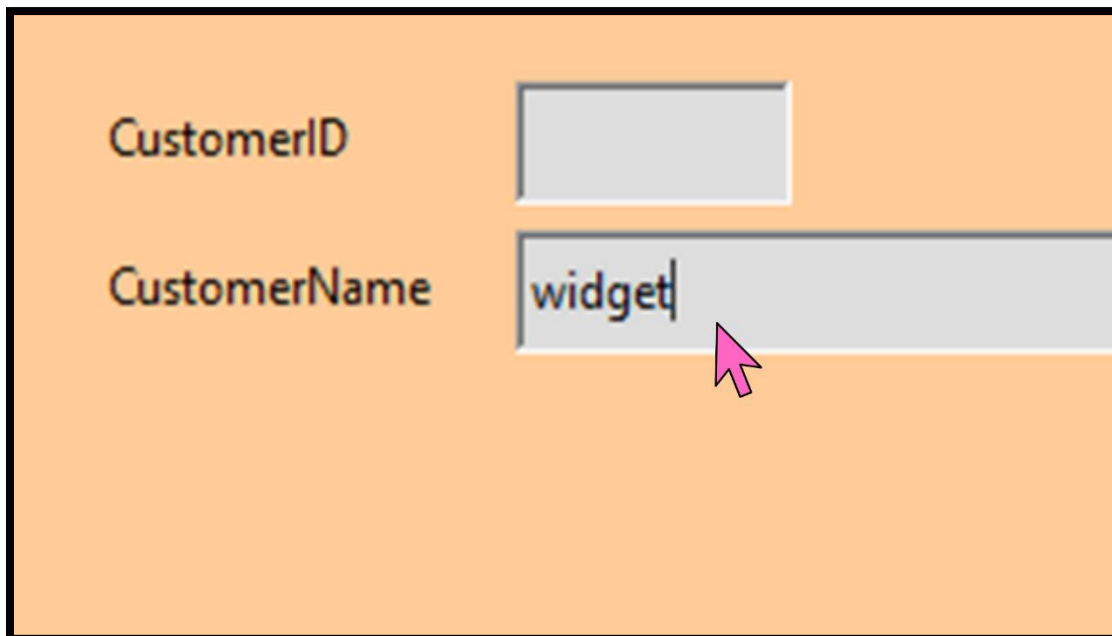




4. In the Forms pane, double-click `CustomerForm`.



5. In the `CustomerForm` window, in the `CustomerName` text box, type `Widget`.





6. In the ContactID column of the subform, type 0.

CustomerID	0
CustomerName	Widget
ContactID	0 

7. In the FirstName column of the subform, type Dave.

ContactID	
Customer	0
FirstName	Dave 
Surname	



8. In the Surname column of the subform, type `Martin`.

ContactID	
Customer	0
FirstName	Dave
Surname	Martin ←

9. In the ContactNum column of the subform, type `5551234567`.

ContactID	5551234567 ←
Customer	0
FirstName	Dave
Surname	Martin



10. In the CustomerName text box, type Grommet.

CustomerID 0

CustomerName Grommet

11. In the ContactID column of the subform, type 0.

ContactID 0

Customer

FirstName

Surname



12. In the FirstName column of the subform, type Sue.

CustomerID	0
CustomerName	Grommet
ContactID	0
Customer	
FirstName	Sue
Surname	

13. In the Surname column of the subform, type Smith.

CustomerID	0
CustomerName	Grommet
ContactID	0
Customer	
FirstName	Sue
Surname	Smith



14. In the ContactNum column of the subform, type 5557654321.

CustomerID: 0

CustomerName: Grommet

ContactID: 5557654321

Customer:

FirstName: Sue

Surname: Smith

15. Click in the subform data sheet and enter one or two more contacts if you wish.

ITFundamentals.odb : Customer 2 - OpenOffice Base: Database Form

File Edit View Insert Format Table Tools Window Help

CustomerID: 0

CustomerName: Grommet

ContactID: 2147483647

Customer:

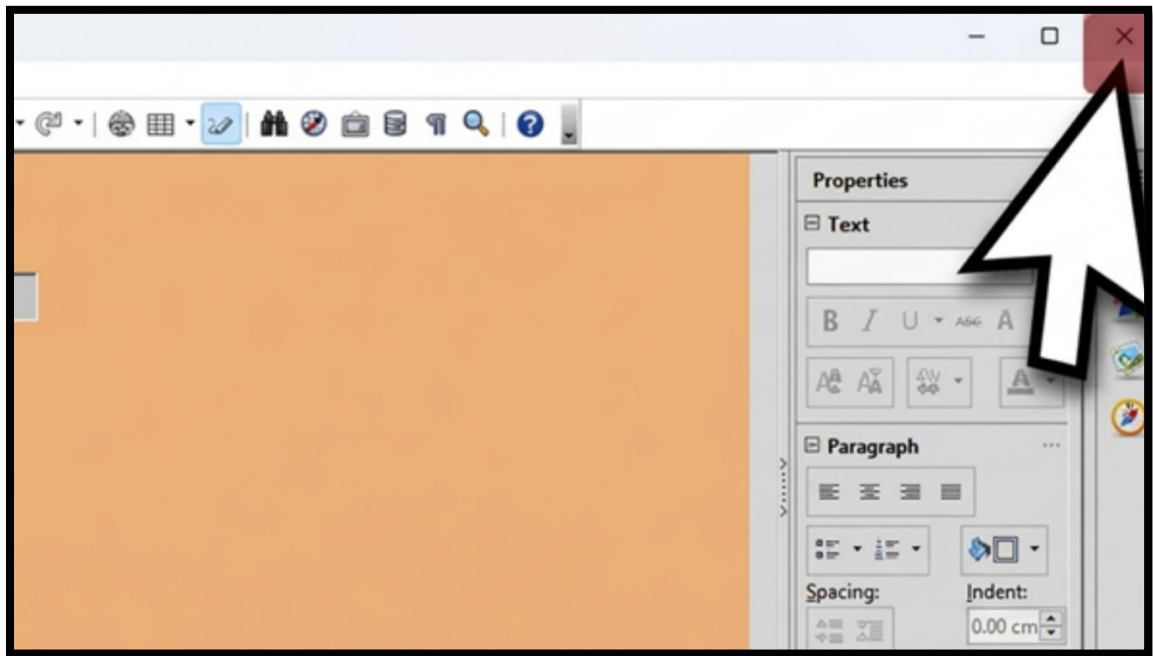
FirstName: Sue

Surname: Smith

Record 1 of 1



16. Close the current form.

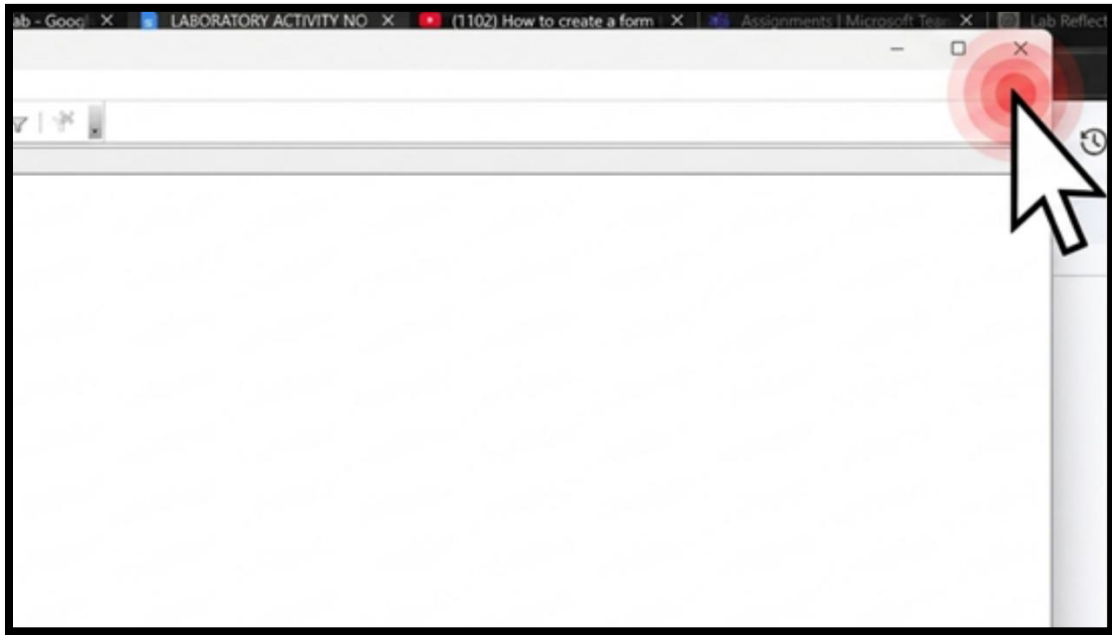


17. In the database window, select the **Tables** object. Open the **Customer** table. Make a note of the values in the CustomerID primary key field for the two records you entered.

	CustomerID	CustomerName
	0	Widget
	1	Grimmet



18. Close the table.

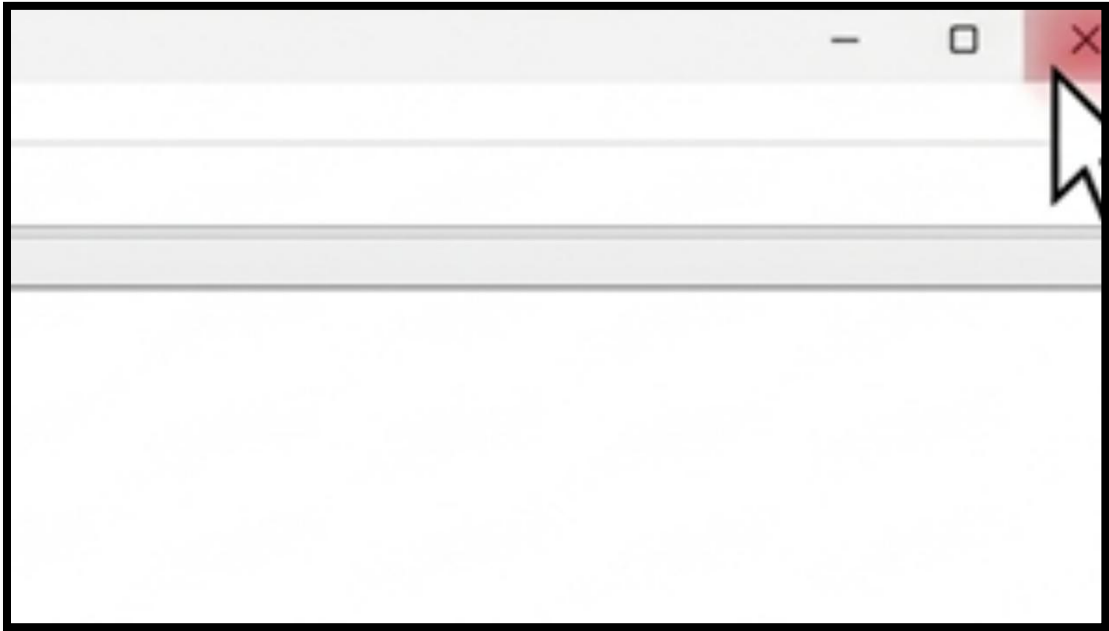


19. Open the **Contact** table and check the Customer field – you should see the values you recorded above entered in the foreign key. These values are added automatically by the database engine because of the relationship you defined between the tables and between the form and subform.

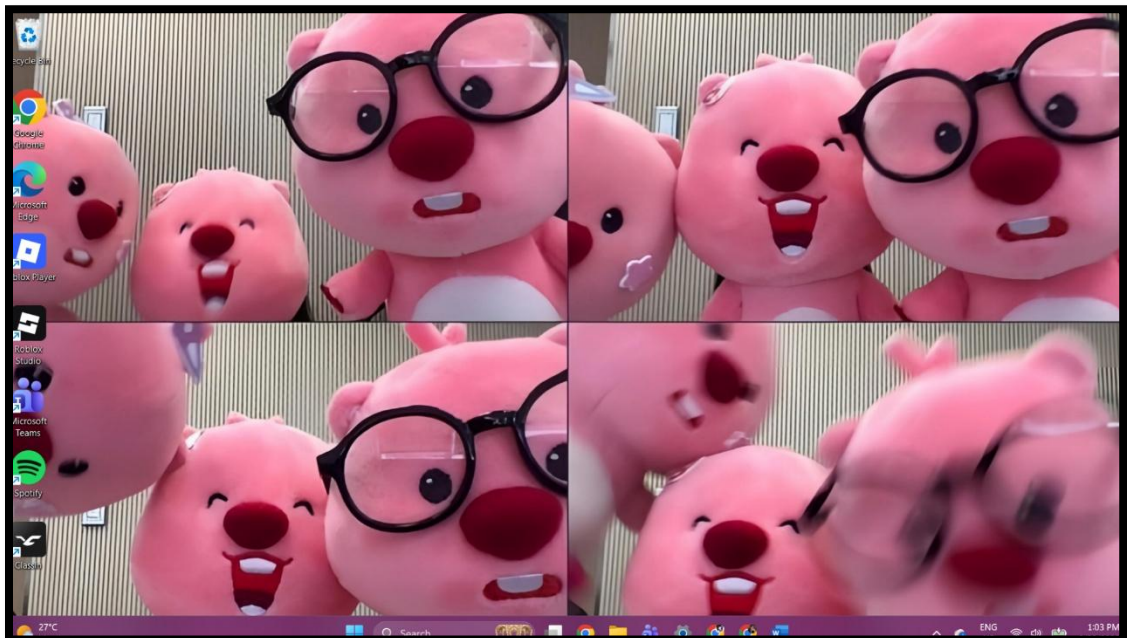
ContactID	Customer	FirstName	Surname
-2147483648	1	Dave	Martin
2147483647	2	Sue	Smith



20. Close the Contact table.



21. Close all open windows and apps.





V. REFLECTION:

Upon successful completion of "Lab 2.4.10: Creating and Using Databases," write a comprehensive reflection (between 150-250 words) addressing the following points:

- **Learning Outcomes:** What new knowledge or skills did you acquire or reinforce regarding the use of a database?
- **Challenges and Solutions:** Describe any difficulties or unexpected issues you encountered during the lab. How did you overcome them, or what steps would you take to resolve them in the future?
- **Real-world Relevance:** How do the skills and understanding gained in this lab relate to real-world scenarios, either in your personal life or a professional IT context?
- **Critical Thinking:** If you had to use an internet image for any step, explain why you couldn't perform the action physically and how finding the image contributed to your understanding.

In this lab, I learned how to create and use databases and how they help organize and manage information. I also improved my skills in following database procedures and troubleshooting simple problems by reviewing the instructions carefully. These skills are useful in real-world IT because databases are commonly used in websites, applications, schools, and businesses. For the steps that required an internet image, I used one because I could not physically perform the action or access the required equipment, and the image helped me understand what the process should look like.